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A DISCREPANCY RESULT FOR HILBERT MODULAR FORMS

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Abstract. Let F be a totally real number field and $r = [F : \mathbb{Q}]$. Let $A_k(\mathfrak{N}, \omega)$ be the space of holomorphic Hilbert cusp forms with respect to $K_1(\mathfrak{N})$, of weight $k = (k_1, \dots, k_r)$ such that $k_j > 2$ for all j, and with central Hecke character ω. For integral ideals $\mathfrak{N}$ and $\mathfrak{n}$ in F such that $(\mathfrak{n}, \mathfrak{N}) = 1$, we study the Petersson trace formula for the Hecke operator $T_{\mathfrak{n}}$ acting on the space $A_k(\mathfrak{N}, \omega)$. We present asymptotic estimates for the terms of the Petersson formula as $k_0 \to \infty$, where $k_0 = \min(k_1, \dots, k_r)$. As an application, we obtain a weighted discrepancy bound for the distribution of the eigenvalues of the Hecke operator $T_{\mathfrak{p}}$ (for a fixed prime ideal $\mathfrak{p}$) acting on the space $A_k(\mathfrak{N}, 1)$, when F has narrow class number 1, and the ideal $\mathfrak{N}$ is generated by (rational) integers. This generalizes a discrepancy result previously obtained by Jung and Sardari [JS20] in the context of classical cusp forms.

1. Introduction

Let F be a totally real number field of degree r over $\mathbb{Q}$. Let $\mathbb{A} = \mathbb{A}_F$ denote the adele ring of F. Let π be a regular, algebraic, cuspidal, automorphic representation for GL_2 over $\mathbb{A}$. Let v be a non-Archimedean place of F where π is unramified. Then, π_v is determined by its Satake parameters $\alpha_\pi(v)$ and $\beta_\pi(v)$. The sum $\kappa_\pi(v) = \alpha_\pi(v) + \beta_\pi(v)$ is the normalised Hecke eigenvalue of the Hecke operator T_v acting on π_v. The generalised Ramanujan conjecture states that $|\alpha_\pi(v)| = |\beta_\pi(v)| = 1$, which is further equivalent to the assertion that $|\kappa_\pi(v)| \leq 2$. This conjecture has been proved in the case when π is holomorphic in a series of important results by [Del74], [DS74], [BL84], [Bla06] and [Ngu05]. We refer the reader to [Li09] for a sequential description of these developments.

In the case when π is not of CM-type, the (generalized) Sato-Tate conjecture states that the Hecke eigenvalues

$$\{\kappa_\pi(v) \mid v \text{ where } \pi_v \text{ is unramified}\}$$

are equidistributed in the interval $[-2, 2]$ with respect to the measure

$$d\mu_\infty(t) := \frac{1}{\pi}\sqrt{1 - \frac{t^2}{4}}dt.$$

Here the representation π is fixed and the primes v vary. This conjecture is now a theorem by the work of [BLGHT11] for the case when $F = \mathbb{Q}$, and [BLGG11] for any totally real field F over $\mathbb{Q}$.

In a related but different perspective, the "vertical" Sato-Tate distribution law considers the equidistribution of these Hecke eigenvalues for a fixed prime v with the representations π varying. To make this precise, fix v and let $\mathfrak{N}$ be an integral ideal of F such that $v \nmid \mathfrak{N}$. Let $k = (k_1, \dots, k_r) \in \mathbb{Z}^r$ with $k_i \geq 2$. Let $\Pi_k(\mathfrak{N})$ denote the set of cuspidal holomorphic automorphic representations with trivial central characters π such that $\pi_{\mathrm{fin}} = \bigotimes_{v \nmid \infty} \pi_v$ has a non-zero $K_0(\mathfrak{N})$-fixed vector and $\pi_\infty = \otimes \pi_{k_i}$, where π_{k_i} are the discrete series representations of weight k_i. In the language of classical Hilbert modular forms, the first condition is equivalent to saying that π comes from a Hilbert modular form of conductor dividing $\mathfrak{N}$. In particular, π_v is unramified for all $v \nmid \mathfrak{N}$. The second condition is equivalent to saying that π comes from a holomorphic Hilbert cusp form of weight $k = (k_1, \dots, k_r)$.

Let $\mathfrak{N}_{\mathfrak{i}}$ be a sequence of ideals (coprime to v) and $\mathfrak{N}_{\mathfrak{i}} \to \infty$ (i.e., the norms of these ideals go to infinity). The vertical Sato-Tate distribution theorem in this case states that the family of multisets

$$S_i = \{\kappa_\pi(v) \mid \pi \in \Pi_k(\mathfrak{N}_i)\}$$

is equidistributed with respect to the measure

$$d\mu_v(x) = \frac{\mathrm{Nm}(v) + 1}{\left(\mathrm{Nm}(v)^{1/2} + \mathrm{Nm}(v)^{-1/2}\right)^2 - x^2} d\mu_\infty(x).$$

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The vertical perspective was first introduced in the work of [Sar87] and [Ser97] in the classical setting of $F = \mathbb{Q}$. While Sarnak [Sar87] addressed the case of Maass cusp forms, Serre [Ser97] studied the vertical framework in the case of holomorphic cusp forms as an application of general equidistribution principles. In the general context of totally real number fields F with $[F : \mathbb{Q}] \geq 2$, the study of the equidistribution of the families S_i was initiated in [Li09].

The equidistribution theorems in the horizontal and vertical aspects are obtained through different techniques. The horizontal Sato-Tate distribution theorem for a fixed non-CM π is obtained by viewing the Hecke eigenvalues $\{\kappa_\pi(v)\}$ as coefficients of the L-function corresponding to π, and showing that the symmetric power L-functions $L(s, \mathrm{Sym}^n \pi)$ for all $n \geq 1$ are potentially cuspidal automorphic. On the other hand, in the vertical context, the families S_i are viewed as the eigenvalues of the Hecke operator $T_{\mathfrak{p}}$ acting on $\Pi_k(\mathfrak{N})$, where $\mathfrak{p}$ is the prime ideal corresponding to the non-Archimedean valuation v. These families are studied with the help of a trace formula for the traces of the Hecke operators $T_{\mathfrak{p}^n}$ for all powers $n \geq 1$. For $F = \mathbb{Q}$, the primary tool is the Eichler-Selberg trace formula. We refer the reader to [KL06] for a detailed description of the history and development of this formula, as well as a proof using the modern perspective of the Arthur-Selberg trace formula for GL_2 over $\mathbb{A}_\mathbb{Q}$. The adelic perspective forms the basis for a suitable generalization of the Eichler-Selberg trace formula to Hilbert modular forms [Li09].

For the purpose of motivating the questions addressed in this article, we start with a review of some equidistribution results in the classical context of modular forms. Let $S_k(N)$ denote the space of cusp forms of even integer weight k with respect to $\Gamma_0(N)$. Let $\mathcal{F}_k(N)$ be an orthonormal basis of $S_k(N)$ consisting of joint eigenfunctions of the Hecke operators T_n with $(n, N) = 1$. For $f \in S_k(N)$, the Fourier expansion of f at the cusp ∞ is given by

$$f(z) = \sum_{n=1}^{\infty} a_f(n) n^{\frac{k-1}{2}} e^{2\pi i n z}.$$

We denote $\kappa_f(n)$ to be the n-th normalised Hecke eigenvalue of f. That is, $\kappa_f(n)$ denotes the eigenvalue of the normalised Hecke operator $T_n/n^{(k-1)/2}$ acting on f. Thus,

$$a_f(n) = a_f(1)\kappa_f(n),\ (n, N) = 1.$$

Let p be a fixed prime number with $\gcd(p, N) = 1$. By the Ramanujan-Deligne bound, we know that $\kappa_f(p) \in [-2, 2]$.

We define, on the interval $[-2, 2]$, the measures

$$d\mu_\infty(x) := \frac{1}{\pi}\sqrt{1 - \frac{x^2}{4}}dx,$$

and

$$d\mu_p(x) := \frac{p+1}{\pi}\frac{1}{(p^{1/2} + p^{-1/2})^2 - x^2}d\mu_\infty(x).$$

For an interval $I \subset [-2, 2]$, and for a fixed prime p, denote

$$\mu_{k,N}(I) := \frac{1}{|\mathcal{F}_k(N)|}\sum_{f \in \mathcal{F}_k(N)} \delta_{\kappa_f(p) \in I},$$

$$\mu_\infty(I) := \int_I d\mu_\infty(x),$$

and

$$\mu_p(I) := \int_I d\mu_p(x).$$

Serre [Ser97] proved that for a fixed prime p, the families of eigenvalues

$$\{\kappa_f(p) \mid f \in \mathcal{F}_k(N)\}$$

are equidistributed in the interval $[-2, 2]$ with respect to the measure $d\mu_p(x)$ as $k + N \to \infty$ with k even and $\gcd(p, N) = 1$. That is, for any interval $I \subset [-2, 2]$,

$$\lim_{\substack{k+N\to\infty \\ (p,N)=1 \\ k \text{ even}}} \mu_{k,N}(I) = \mu_p(I). \tag{1}$$

The above asymptotic for the case $N = 1$ was also proved by Conrey, Duke and Farmer [CDF97].

Let $S_k(N)^*$ be the subspace of primitive cusp forms in $S_k(N)$. Let T_n^* be the restriction of Hecke operator T_n on the subspace $S_k(N)^*$ of $S_k(N)$. Let $\mathcal{F}_k(N)^*$ be the orthogonal basis of $S_k(N)^*$ consisting of joint eigenfunctions of the Hecke operators T_n^*, $(n, N) = 1$ such that $a_f(1) = 1$ for all $f \in \mathcal{F}_k(N)^*$.

We denote $\mu^*_{k,N}$ to be the corresponding measure on $[-2,2]$ associated to T^*_p. That is, for an interval $I \subset [-2,2]$,

$$\mu^*_{k,N}(I) := \frac{1}{|\mathcal{F}_k(N)^*|} \sum_{f \in \mathcal{F}_k(N)^*} \delta_{\kappa_f(p) \in I}.$$

Using (1), it is not too difficult to deduce that for a fixed prime p, and for an interval $I \subset [-2,2]$,

$$\lim_{\substack{k+N \to \infty \\ (p,N)=1 \\ k \text{ even}}} \mu^*_{k,N}(I) = \mu_p(I). \tag{2}$$

The above mentioned equidistribution theorems naturally lead to questions about the discrepancies

$$D(\mu_{k,N}, \mu_p) := \sup \left\{ |\mu_{k,N}(I) - \mu_p(I)| \,\middle|\, I = [a,b] \subset [-2,2] \right\}$$

and

$$D(\mu^*_{k,N}, \mu_p) := \sup \left\{ |\mu^*_{k,N}(I) - \mu_p(I)| \,\middle|\, I = [a,b] \subset [-2,2] \right\}.$$

These have been well investigated, along with applications (see [GJS99], [Roy00], [Gol04], [MS09], [MS10], [SZ24], [JS20]).

In [GJS99], Gamburd, Jakobson and Sarnak obtain the bound

$$D(\mu_{k,2}, \mu_p) = \mathrm{O}\left(\frac{1}{\log k}\right)$$

by a study of the eigenvalues of $\widehat{z}(\pi_k)$, as $k \to \infty$, where z is an element in the group ring $\mathbb{R}[\mathrm{SU}(2)]$, and π_k is the irreducible representation of $\mathrm{SU}(2)$ of dimension $k+1$ (see [GJS99, Theorem 1.3]).

For $N = 1$, Golubeva [Gol04] obtains the bound

$$D(\mu_{k,1}, \mu_p) = \mathrm{O}_\epsilon\left(\frac{1}{(\log k)^{1-\epsilon}}\right)$$

for any $\epsilon > 0$.

This can be sharpened into the bound

$$D(\mu_{k,N}, \mu_p) = \mathrm{O}\left(\frac{1}{\log kN}\right) \tag{3}$$

[MS09], which holds for all even positive integers k and for all $N \geq 1$ such that $(p,N) = 1$. A similar upper bound [MS10] can also be obtained for $D(\mu^*_{k,N}, \mu_p)$:

$$D(\mu^*_{k,N}, \mu_p) = \mathrm{O}\left(\frac{1}{\log kN}\right). \tag{4}$$

The primary tool used in [Gol04] and [MS09] is the Eichler-Selberg trace formula, combined with an approximation of the characteristic function of an interval by certain smooth test functions. Indeed, for any interval $I \subset [-2,2]$, and any positive integer $M \geq 1$, we have an Erdös-Turán type of inequality:

$$|\mu_{k,N}(I) - \mu_p(I)| \leq \frac{b_1 \dim(S_k(N))}{M+1} + \sum_{m=1}^{M} b_{m,I} G(m), \tag{5}$$

where b_1 and $b_{m,I}$ are positive constants derived from the above test functions, and $G(m)$ depends on the sum $\sum_{f \in \mathcal{F}_k(N)} \kappa_f(p^m)$, which is $\mathrm{Tr}\left(\frac{T_{p^m}}{p^{m(k-1)/2}}\right)$. We choose a value of M to obtain an optimal bound for the right hand side of (5). This technique works for all positive integers N (coprime to p), and gives us the discrepancy bounds (3) and (4), where the implied constants are absolute and effectively computable. However, this method faces a barrier: the Eichler-Selberg trace-formula estimates for the sum $\sum_{f \in \mathcal{F}_k(N)} \kappa_f(p^m)$, when applied to (5), generate a non-trivial bound only for choices of M such that $p^M \ll (kN)^c$ for some $0 < c < 2/3$.

The application of the Petersson trace formula for the study of discrepancies has been explored in [JS20] and [SZ24]. This formula evaluates the harmonic sums

$$\sum_{f \in \mathcal{F}_k(N)} \omega_f \kappa_f(n_1) \overline{\kappa_f(n_2)},$$

where

$$\omega_f := \frac{\Gamma(k-1)}{(4\pi)^{k-1}} |a_f(1)|^2, \tag{6}$$

and n_1 and n_2 are positive integers coprime to N. For an application to the current context, we choose $n_1 = p^m$ and $n_2 = 1$. The perspective of [SZ24] is to extend the range of m for a sharper estimation of

the trace $\sum_{f\in\mathcal{F}_k(N)}\kappa_f(p^m)$ through the Petersson trace formula. A weight-removal technique is applied to the above sum by interpreting the weight ω_f in terms of the special value of the symmetric square L-function $L(s,\mathrm{Sym}^2 f)$ at $s=1$, and by a suitable approximation for $L(s,\mathrm{Sym}^2 f)$. This approach was followed in [SZ24] to derive bounds for the discrepancies $D(\mu_{k,N},\mu_p)$ and $D(\mu^*_{k,N},\mu_p)$ in the case when $k=2$. In the context of Hecke-Maass forms, the use of the analogous Kuznetsov trace formula and weight removal to obtain the requisite discrepancy bounds have been explored in [LW11] and [TW16].

Question. *Is it possible to improve the logarithmic saving in* (3) *to a power saving in the bound for* $D(\mu_{k,N},\mu_p)$ *(and likewise, for* $D(\mu^*_{k,N},\mu_p)$*)? That is, does there exist* $0<\theta<1$ *such that*

$$D(\mu_{k,N},\mu_p)=\mathrm{O}\left(\frac{1}{(kN)^{\theta}}\right)?$$

Related to the above question is that of obtaining Ω-type estimates for these discrepancies. In what follows, we restrict our attention to the spaces $S_k(N)^*$.

Question. *Can we find a function* $E(k,N)$ *for positive integers* (k,N) *such that*

$$D(\mu^*_{k,N},\mu_p)=\Omega(E(k,N))\ \text{ as } k+N\to\infty?$$

That is, can we find a sequence $(k_n,N_n)_{n\geq 1}$ *such that* k_n *is even,* $(p,N_n)=1$ *and*

$$D(\mu^*_{k_n,N_n},\mu_p)\gg E\left(k_n,N_n\right)\ \text{ as } k_n+N_n\to\infty?$$

The above question was addressed for $N=2$ by Gamburd, Jakobson and Sarnak [GJS99], and for all squarefree levels N by Jung and Sardari [JS20]. In [GJS99], it is shown that there exists a sequence of even integers $k_n\to\infty$ such that

$$D(\mu^*_{k_n,2},\mu_p)\gg\frac{1}{k_n^{\frac{1}{2}}\log^2 k_n}. \tag{7}$$

Jung and Sardari [JS20, Theorem 1.6] generalize the above result to any fixed squarefree level N with an improved exponent for k_n. That is, given a fixed squarefree level N, they obtain a sequence of weights k_n with $k_n\to\infty$ such that

$$D(\mu^*_{k_n,N},\mu_p)\gg\frac{1}{k_n^{\frac{1}{3}}\log^2 k_n}. \tag{8}$$

This tells us that if the answer to the first question above is affirmative, then $\theta<1/3$.

Define

$$\mathcal{H}_k(N)^*:=\sum_{f\in\mathcal{F}_k(N)^*}\omega_f,$$

with ω_f as in (6).

We define the Borel measure $\nu^*_{k,N}$ on $[-2,2]$ as follows: for an interval $I\subset[-2,2]$,

$$\nu^*_{k,N}(I):=\frac{1}{\mathcal{H}_k(N)^*}\sum_{f\in\mathcal{F}^*_k(N)}\omega_f\,\delta_{\kappa_f(p)\in I}.$$

Using the Petersson trace formula, one can show a weighted variant of (2), namely that

$$\lim_{\substack{k+N\to\infty\\(p,N)=1\\k\text{ even}}}\nu^*_{k,N}(I)=\mu_\infty(I). \tag{9}$$

We refer the interested reader to [Li04], [OM09] and [KL13] for a discussion of weighted distribution theorems for Hecke eigenvalues, other variants and analogues for Hecke-Maass cusp forms.

In the context of discrepancies, Jung and Sardari obtain, for a fixed squarefree level N, a sequence of weights k_n with $k_n\to\infty$ such that the lower bound

$$D(\nu^*_{k_n,N},\mu_\infty)\gg\frac{1}{k_n^{\frac{1}{3}}\log^2 k_n} \tag{10}$$

holds. The transition from (10) to (8) is made with the help of an explicit asymptotic version of the Petersson trace formula. Recently, an extension of (10) to any positive integer N which is not divisible by 8 was obtained in [Das24, Theorem 1].

The discrepancy bound in (3) was extended to the context of Hilbert modular forms in [LLW14]. This gives us a quantitative version of the vertical equidistribution result of [Li09] with effective error terms. The strategy of proof in [LLW14] is to use a "higher-dimensional" variant of the Erdös-Turan inequality (see [Li09, Section 7]). The analogue of the term $G(m)$ in the case of Hilbert modular forms is then estimated with the help of the Arthur-Selberg trace formula (see [LLW14, Section 3]).

However, the extension of the Ω-type bounds in (8) and (10) to Hilbert modular forms has not been addressed yet. In this article, we initiate a discussion on the extension of (10) to Hilbert modular forms. We first recall the following weighted equidistribution result of Knightly and Li [KL08, Theorem 1.1].

Theorem ([KL08])**.** *Let F be a totally real number field, and let m be a totally positive element of the inverse different $\mathfrak{d}^{-1} \subset F$. For a cusp form ϕ on $\mathrm{GL}_2(F)\backslash\mathrm{GL}_2(\mathbb{A}_F)$ with trivial central character, let W_m^ϕ denote its m-th Fourier coefficient (see (19)). The weight associated to the cusp form ϕ is defined as*

$$w_\phi := \frac{|W_m^\phi(1)|^2}{\|\phi\|^2},$$

where $\|\phi\|$ is the Petersson norm of ϕ.

For an integral ideal $\mathfrak{N}$ of $\mathcal{O}$, let $A_k(\mathfrak{N}, 1)$ denote the space of holomorphic Hilbert cusp forms of weight $k = (k_1, k_2, \ldots, k_r)$ (each $k_i > 2$ and even) with respect to the Hecke congruence subgroup $\Gamma_0(\mathfrak{N})$.

For integral ideals $\mathfrak{n}$ and $\mathfrak{N}$ such that $(\mathfrak{n}, \mathfrak{N}) = 1$, let $T_\mathfrak{n}$ denote the $\mathfrak{n}$-th Hecke operator acting on $A_k(\mathfrak{N}, 1)$. Let $\mathcal{F}_k(\mathfrak{N})$ be a Hecke eigenbasis of $A_k(\mathfrak{N}, 1)$, that is, a basis of $A_k(\mathfrak{N}, 1)$ consisting of simultaneous eigenfunctions of the Hecke operators $T_\mathfrak{n}$.

For each $\phi \in \mathcal{F}_k(\mathfrak{N})$, let $\lambda_\mathfrak{n}^\phi$ be an eigenvalue for the Hecke operator $T_\mathfrak{n}$ with eigenvector ϕ. Let $\kappa_\mathfrak{n}^\phi := \frac{\lambda_\mathfrak{n}^\phi}{\sqrt{\mathrm{Nm}(\mathfrak{n})}}$.

Consider a fixed prime ideal $\mathfrak{p}$ not dividing $m\mathfrak{d}$. For an interval $I \subset [-2, 2]$, define

$$\nu_{k,\mathfrak{N}}(I) := \frac{1}{\mathcal{H}_k(\mathfrak{N})} \sum_{\phi \in \mathcal{F}_k(\mathfrak{N})} \omega_\phi \delta_{\kappa_\mathfrak{p}^\phi \in I}, \tag{11}$$

where

$$\mathcal{H}_k(\mathfrak{N}) = \sum_{\phi \in \mathcal{F}_k(\mathfrak{N})} \omega_\phi.$$

Then,

$$\lim_{\substack{\mathrm{Nm}(\mathfrak{N}) \to \infty \\ (\mathfrak{p}, \mathfrak{N}) = 1}} \nu_{k,\mathfrak{N}}(I) = \mu_\infty(I).$$

This leads to questions about the discrepancies $D(\nu_{k,\mathfrak{N}}, \mu_\infty)$. In the current article, our main goal is to generalize equation (10) to the context of Hilbert cusp forms. The proof of (10) follows from asymptotic estimates for the Petersson trace formula for the Hecke operators T_n acting on the space $S_k(N)^*$ where the level N is square-free, and n lies in a certain range dependent on k. Thus, the first theorem in this article is an asymptotic version of the Petersson trace formula for Hilbert cusp forms. We review some notation before the statement of the theorem.

- Let F^+ denote the set of totally positive elements of F. Let $\mathbb{A}$ (or $\mathbb{A}_F$) denote the adele ring of F. Since F is fixed throughout this manuscript, we usually omit it from the notation.
- Let $\mathcal{O}^\times$ denote the unit group of F, and let U be a fixed set of representatives for $\mathcal{O}^\times/\mathcal{O}^{\times 2}$. Note that U is a finite set and $|U| = 2^r$, where $[F : \mathbb{Q}] = r$.
- Let $\sigma_1, \ldots, \sigma_r$ be the embeddings of F into $\mathbb{R}$ and let $\sigma = (\sigma_1, \ldots, \sigma_r) : F \to \mathbb{R}^r$.
- Let $\mathfrak{n}$ and $\mathfrak{N}$ be ideals in $\mathcal{O}$ such that $(\mathfrak{n}, \mathfrak{N}) = 1$.
- For a non-Archimedean place v of F, let $\mathcal{O}_v$ denote the ring of integers in the local field F_v. Let $\widehat{\mathcal{O}} = \prod_{v<\infty} \mathcal{O}_v$.
- For a fractional ideal $\mathfrak{a} \subset F$, let $\mathfrak{a}_v$ denote its localization. We write

  $$\widehat{\mathfrak{a}} = \mathfrak{a}\widehat{\mathcal{O}} = \prod_{v<\infty} \mathfrak{a}_v.$$

  Further, let $[\mathfrak{a}]$ denote the image of $\mathfrak{a}$ in the ideal class group.
- Consider the equation $1 = [\mathfrak{b}]^2[\mathfrak{n}]$ in terms of ideal class groups, and let

  $$[\mathfrak{b}_1],\ [\mathfrak{b}_2],\ \ldots,\ [\mathfrak{b}_t]$$

  be solutions of the above equation. We choose $\eta_i \in F$ such that η_i generates the principal ideal $\mathfrak{b}_i^2\mathfrak{n}$. Also, $\mathfrak{b}_i$ is such that $\mathfrak{b}_i\hat{\mathcal{O}} = \hat{\mathfrak{b}}_i$ for the integral ideal $\mathfrak{b}_i$.
- Let $\|x\|$ denote the standard Euclidean norm of $x \in \mathbb{R}^r$. Let us consider

  $$\delta_i = \inf\{\|\sigma(s)\| \mid s \in \mathfrak{b}_\mathfrak{i}\mathfrak{N}/\pm, s \neq 0\}, \tag{12}$$

  where each $\mathfrak{b}_\mathfrak{i}$ is as defined above. We also denote $\tilde{\delta}_i = \frac{\delta_i}{2\sqrt{r}}$.

- We define
$$A_i = \bigcap_{j=1}^{r} \{s \in \mathfrak{b}_i\mathfrak{N}/\pm \mid |\sigma_j(s)| \leq 2\delta_i,\ s \neq 0\}. \tag{13}$$
For each i, A_i is a discrete bounded set in $\mathbb{R}^r$, hence finite. This will contribute to the main term in our estimates below.
- We define
$$\gamma_j = \max\left\{\sqrt{\sigma_j(\eta_i u)} \mid i = 1, \ldots, t, u \in U, \eta_i u \in F^+\right\}.$$

Theorem 1. *Let $\mathfrak{N}$ and $\mathfrak{n}$ be fixed integral ideals in F such that $(\mathfrak{n}, \mathfrak{N}) = 1$. Let $\omega : F^\times\backslash\mathbb{A}^\times \to \mathbb{C}^\times$ be a unitary Hecke character. Let the conductor of ω divide $\mathfrak{N}$ and $\omega_{\infty_j}(x) = \mathrm{sgn}(x)^{k_j}$ for all $j = 1, \ldots, r$. For $k = (k_1, k_2, \ldots, k_r)$ with all $k_j > 2$, let $A_k(\mathfrak{N}, \omega)$ denote the space of Hilbert cusp forms of weight k and character ω with respect to $K_1(\mathfrak{N})$ (see equation (18) for the definition of $K_1(\mathfrak{N})$) and let $\mathcal{F}$ be an orthogonal basis for $A_k(\mathfrak{N}, \omega)$ consisting of eigenfunctions of the Hecke operator $T_\mathfrak{n}$. Let A_i's and $\tilde{\delta}_i$ be as defined above and $k_0 = \min\{k_j \mid j \leq r\}$. Suppose $m_1, m_2 \in \mathfrak{d}_+^{-1}$ and satisfy*
$$\frac{2\pi\gamma_j\sqrt{\sigma_j(m_1m_2)}}{\tilde{\delta}_i} \in \Big((k_j-1) - (k_j-1)^{\frac{1}{3}}, (k_j-1)\Big) \text{ for all } j \leq r. \tag{14}$$
Then, as $k_0 \to \infty$,
$$\frac{e^{2\pi \mathrm{Tr}_{\mathbb{Q}}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\left[\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi\sqrt{\sigma_j(m_1m_2)})^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}}\frac{\lambda_\mathfrak{n}^\phi W_{m_1}^\phi(1)\overline{W_{m_2}^\phi(1)}}{\|\phi\|^2}$$
$$= \hat{T}(m_1, m_2, \mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_\mathfrak{N}(m_1/s)\omega_\mathrm{f}(s)} + \sum_{i=1}^{t}\sum_{u\in U, \eta_i u\in F^+}\sum_{s\in A_i}\Bigg\{\omega_\mathrm{f}(s\mathfrak{b}_i^{-1})S_{\omega_\mathfrak{N}}(m_1, m_2; \eta_i u\mathfrak{b}_i^{-2}; s\mathfrak{b}_i^{-1})$$
$$\frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)}\times\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\Bigg\} + \mathrm{O}_{A_i,u,\eta_i}\left(\sum_{j=1}^{r}\left((k_j-1)^{-\frac{1}{2}}\prod_{\substack{j'=1\\ j'\neq j}}^{r}(k_{j'}-1)^{-\frac{1}{3}}\right)\right),$$
for all j. A detailed description of all the terms above is provided in §2.

Remark 1. Here are some features and comments for the above theorem.

- We note that the values of m_1 and m_2 are dependent on k. Since the lengths of the intervals $\Big((k_j-1) - (k_j-1)^{\frac{1}{3}}, (k_j-1)\Big)$ go to infinity, we can always choose infinitely many m_1, m_2 satisfying (14), as $k_0 \to \infty$.
- The main idea of the proof is to use the Petersson trace formula in the Hilbert modular setting, which was first stated in [KL08] and is recalled in this article as Theorem 5. We then apply Lemma 8 carefully using a combinatorial arithmetic argument to arrive at the estimate.
- Theorem 1 is an analogue of [JS20, Theorem 1.7] for the space $A_k(\mathfrak{N}, \omega)$. See also [Das24, Theorem 4(i)].

In our next result, we find a lower bound for the main term in Theorem 1 under additional assumptions that F has odd narrow class number and the ideal $\mathfrak{b}_1\mathfrak{N} = \tilde{s}\mathcal{O}$, for $\tilde{s} \in \mathbb{Z}$. For details about notations in the next theorem, see §2.

Theorem 2. *We continue with the notations and assumptions of Theorem 1. In addition, let F have odd narrow class number. Further let $\mathfrak{b}_1\mathfrak{N} = \tilde{s}\mathcal{O}$ with $\tilde{s} \in \mathbb{Z}$. The following statements are true.*

(a) *The main term in Theorem 1 is equal to*
$$\hat{T}(m_1, m_2, \mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_\mathfrak{N}(m_1/s)\omega_\mathrm{f}(s)} + \Bigg\{\omega_\mathrm{f}(s\mathfrak{b}_1^{-1})S_{\omega_\mathfrak{N}}(m_1, m_2; \eta_1\mathfrak{b}_1^{-2}; s\mathfrak{b}_1^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_1)}}{\mathrm{Nm}(s)}$$
$$\times\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_1 m_1 m_2)}}{|\sigma_j(s)|}\Big)\Bigg\}$$
where $s = |\tilde{s}|$.

(b) *Further assume that* $S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_1\mathfrak{b}_1^{-2};s\mathfrak{b}_1^{-1})\neq 0$ *for all* $m_1,m_2\in\mathfrak{d}_+^{-1}$. *Then,*

$$\left|\frac{e^{2\pi tr_{\mathbb{Q}}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi\sqrt{|\sigma_j(m_1m_2)|})^{k_j-1}}\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{n}}^{\phi}W_{m_1}^{\phi}(1)\overline{W_{m_2}^{\phi}(1)}}{\|\phi\|^2}\right.$$
$$\left.-\hat{T}(m_1,m_2,\mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_{\mathfrak{N}}(m_1/s)\omega_{\mathfrak{f}}(s)}\right|\gg\prod_{j=1}^{r}(k_j-1)^{-\frac{1}{3}},$$

as $k_0\to\infty$.

Remark 2. For squarefree N, the main term in [JS20, Theorem 1.7] is

$$2\pi i^{-k}\frac{\mu(N)}{N}\prod_{p|N}(1-p^{-2})J_{k-1}(4\pi\sqrt{mn}).$$

Note that $\mu(N)=S(1,0,N)$, where $S(a,b,c)$ denotes the classical Kloosterman sum, which is nonzero if and only if N is squarefree.

In our context, we show in Lemma 13 that for some specific values of $\eta_1,\mathfrak{b}_1$ and s and the trivial character $\omega_{\mathfrak{N}}$, the Kloosterman sum $S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_1\mathfrak{b}_1^{-2};s\mathfrak{b}_1^{-1})\neq 0$ for all $m_1,m_2\in\mathfrak{d}_+^{-1}$.

The main idea behind the proof of Theorem 2 is to reduce the triple sum

$$\sum_{i=1}^{t}\sum_{u\in U,\eta_iu\in F^+}\sum_{s\in A_i}$$

in Theorem 1 into a single term where finding a lower bound is possible. Similar lower bounds are obtained in [Das26] when F is a multi-quadratic field.

Now we consider the following application of Theorem 2. An analogue of $\nu_{k,N}$ for Hilbert cusp forms can be defined in the following manner. Let

$$\tilde{\nu}_{k,\mathfrak{N}}:=\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi)^{k_j-1}}\sum_{\phi\in\mathcal{F}}\frac{\delta_{\kappa_{\mathfrak{p}}^{\phi}}}{\|\phi\|^2},\tag{15}$$

where $\mathcal{F}$ is an orthogonal basis for $A_k(\mathfrak{N},\omega)$ consisting of eigenfunctions of the Hecke operator $T_{\mathfrak{p}}$ with $(\mathfrak{N},\mathfrak{p})=1$.

Remark 3. Note that the measure considered in (15) is a variant of the measure considered (11). To see the choices for m_1,m_2 and $\mathfrak{n}$ that were made to obtain $\tilde{\nu}_{k,\mathfrak{N}}$, the reader is referred to §5.

We obtain the following variant of (10) ([JS20, Theorem 1.6]) for the space $A_k(\mathfrak{N},1)$.

Theorem 3. *We continue with the notations and assumptions of Theorem 1. Let* F *have narrow class number* 1. *Let* $\mathfrak{b}_1\mathfrak{N}=\tilde{s}\mathcal{O}$ *with* $\tilde{s}\in\mathbb{Z}$ *such that* $|\tilde{s}|$ *is squarefree and* $\omega_{\mathfrak{N}}$ *is trivial. Let* $\mathfrak{p}=\tilde{p}\mathcal{O}$ *be a prime ideal such that* $|\sigma_j(\tilde{p})|>1$ *for all* $j=1,\dots,r$. *There exists an infinite sequence of weights* $k_l=(k_{l_1},...,k_{l_r})$ *with* $(k_l)_0\to\infty$ *such that*

$$D(\tilde{\nu}_{k_l,\mathfrak{N}},\mu_\infty)\gg\frac{1}{\big(\log(k_l)_0\big)^2\times\prod_{i=1}^{r}(k_{l_i}-1)^{\frac{1}{3}}}$$

as $l\to\infty$.

Remark 4.
- It is conjectured that there are infinitely many real quadratic fields with narrow class number 1. Some examples of such fields are given in the OEIS sequence A003655.
- The assumption that $|\sigma_j(\tilde{p})|>1$ for all $j=1,\dots,r$ is satisfied for a rational prime $\tilde{p}$ inert in F.
- In the above Theorem, we only prove the discrepancy result for the weighted measure $\tilde{\nu}_{k_l,\mathfrak{N}}$. We expect a similar result for the unweighted measure as well (similar to [JS20]), which we hope to prove in future work.

Organisation of this article. In §2, we set up some basic notations and definitions for Hilbert modular forms. We also recall the Petersson trace formula for these forms and a related estimate from [KL08]. In §3, we prove Theorem 1, which is an estimate of the error term in the Petersson trace formula. In §4, we analyse the main term from Theorem 1 in special cases and prove Theorem 2. Finally, in §5, we prove Theorem 3 which gives an Ω-estimate for the discrepancy between the measures $\tilde{\nu}_{k,\mathfrak{N}}$ and μ_∞.

2. Petersson trace formula

In this section, we recall some basic facts about Hilbert modular forms and the Petersson trace formula for the space $A_k(\mathfrak{N}, \omega)$.

Let F be a totally real number field and $r = [F : \mathbb{Q}]$. Let $\sigma_1, \dots, \sigma_r$ be the distinct embeddings of $F \hookrightarrow \mathbb{R}$. Let $\sigma : F \to \mathbb{R}^r$ be given by $\sigma(s) = (\sigma_1(s), \dots, \sigma_r(s))$. Let $\infty_1, \dots, \infty_r$ denote the corresponding Archimedean valuations. Let $\mathcal{O}$ be the ring of integers of F. Let $\mathrm{Nm} : F \to \mathbb{Q}$ denote the norm map. For a nonzero ideal $\mathfrak{a} \subset \mathcal{O}$, let $\mathrm{Nm}(\mathfrak{a}) = |\mathcal{O}/\mathfrak{a}|$. For $\alpha \in F^\times$, we have $\mathrm{Nm}(\alpha\mathcal{O}) = |\mathrm{Nm}(\alpha)|$. Let d_F denote the discriminant of F.

Let $\nu = \nu_{\mathfrak{p}}$ be the discrete valuation corresponding to a prime ideal $\mathfrak{p}$. Let F_ν be the completion of F with respect to the valuation ν. Let $\mathcal{O}_\nu$ be the ring of integers of the local field F_ν. Let $\mathbb{A}$ denote the adèle ring of F with finite adèles $\mathbb{A}_f$, so that $\mathbb{A} = F_\infty \times \mathbb{A}_f$ where $F_\infty = F \otimes \mathbb{R} \cong \mathbb{R}^r$. Let $\hat{\mathcal{O}} = \prod_{\nu<\infty} \mathcal{O}_\nu \subset \mathbb{A}_f$. Let F^+ denote the set of totally positive elements of F, i.e., all $x \in F$ such that $\sigma_i(x) > 0$ for all $i = 1, \dots, r$. We let F_∞^+ denote the subset of F_∞ consisting of vectors whose entries are all positive. Let $\mathfrak{d}^{-1} = \{x \in F : \mathrm{Tr}_{\mathbb{Q}}^F(x\mathcal{O}) \subset \mathbb{Z}\}$ denote the inverse different ideal. We also let $\mathfrak{d}_+^{-1} = \mathfrak{d}^{-1} \cap F^+$.

Let $\mathfrak{N}$ be an integral ideal of $\mathcal{O}$. Let $k = (k_1, \dots, k_r)$ be an r-tuple of integers with $k_j \geq 2$. Let $\omega : F^\times\backslash\mathbb{A}^\times \to \mathbb{C}^\times$ be a unitary Hecke character. We can decompose ω as $\omega = \prod_\nu \omega_\nu$, where $\omega_\nu : F_\nu^\times \to \mathbb{C}^\times$ are the local characters. We further assume that

(1) the conductor of ω divides $\mathfrak{N}$,
(2) $\omega_{\infty_j}(x) = \mathrm{sgn}(x)^{k_j}$ for all $j = 1, \dots, r$.

The first condition means that ω_ν is trivial on $1 + \mathfrak{N}\mathcal{O}_\nu$ for all $\nu|\mathfrak{N}$, and unramified for all $\nu \nmid \mathfrak{N}$.

Let $\theta : \mathbb{A} \to \mathbb{C}^\times$ be the standard character of $\mathbb{A}$. Concretely, $\theta(x) = \theta_\infty(x_\infty) \cdot \prod_{\nu<\infty} \theta_\nu(x_\nu)$, where

(1) $\theta_\infty : F_\infty \to \mathbb{C}^\times$ is defined by $\theta_\infty(x_\infty) = e^{-2\pi i(x_1 + \cdots + x_r)}$ for $x_\infty = (x_1, \dots, x_r)$, and
(2) for $\nu < \infty$, $\theta_\nu : F_\nu \to \mathbb{C}^\times$ is given by $\theta_\nu(x_\nu) = e^{2\pi i\{\mathrm{Tr}_\nu(x_\nu)\}}$. Here $\{\mathrm{Tr}_\nu(x_\nu)\}$ is obtained by composing the following maps: $\mathrm{Tr}_{\mathbb{Q}_p}^{F_\nu} : F_\nu \to \mathbb{Q}_p$, going modulo p-adic integers: $\mathbb{Q}_p \to \mathbb{Z}_p$, and identifying $\mathbb{Q}_p/\mathbb{Z}_p$ with $\mathbb{Q}/\mathbb{Z}$. This map is well-defined since $e^{2\pi i\mathbb{Z}} = 1$. Moreover, the kernel of θ_ν is the local inverse different $\mathfrak{d}_\nu^{-1} = \{x \in F_\nu : \mathrm{Tr}_{\mathbb{Q}_p}^{F_\nu}(x) \in \mathbb{Z}_p\}$.

Note that $\theta(x) = 1$ for $x \in F$, therefore θ is a character on $F\backslash\mathbb{A}$.

We now define Kloosterman sums, first locally and then globally. For any finite valuation ν of F, let $\mathfrak{n}_\nu \in \mathcal{O}_\nu$, $\mathfrak{n}_\nu \neq 0$, and $\mathfrak{m}_{1\nu}, \mathfrak{m}_{2\nu} \in \mathfrak{d}_\nu^{-1}$. For $\mathfrak{c}_\nu \in \mathfrak{N}_\nu$, $\mathfrak{c}_\nu \neq 0$, we define the local Kloosterman sum

$$S_{\omega_\nu}(\mathfrak{m}_{1\nu}, \mathfrak{m}_{2\nu}; \mathfrak{n}_\nu; \mathfrak{c}_\nu) = \sum_{\substack{s_1, s_2 \in \mathcal{O}_\nu/\mathfrak{c}_\nu\mathcal{O}_\nu \\ s_1 s_2 \equiv \mathfrak{n}_\nu \bmod c_\nu\mathcal{O}_\nu}} \theta_\nu\Big(\frac{\mathfrak{m}_{1\nu}s_1 + \mathfrak{m}_{2\nu}s_2}{\mathfrak{c}_\nu}\Big)\omega_\nu(s_2)^{-1}. \tag{16}$$

The value of the sum is 1 if $\mathfrak{c}_\nu \in \mathcal{O}_\nu^\times$.

For a fractional ideal $\mathfrak{a}$, let $\hat{\mathfrak{a}} = \prod_{\nu<\infty} \mathfrak{a}_\nu \subset \mathbb{A}_f$ where $\mathfrak{a}_\nu$ denotes its localisation. Let

$$\theta_f(x) = \prod_{\nu<\infty} \theta_\nu(x_\nu),$$

and let

$$\omega_{\mathfrak{N}} = \prod \omega_{\mathfrak{N},\nu} = \prod_{\nu|\mathfrak{N}} \omega_\nu.$$

where $\omega_{\mathfrak{N},\nu} = \omega_\nu$ if $\nu|\mathfrak{N}$, and 1 if $\nu \nmid \mathfrak{N}$. For $\mathfrak{n} \in \hat{\mathcal{O}} \cap \mathbb{A}_f^\times$, $\mathfrak{c} \in \hat{\mathfrak{N}} \cap \mathbb{A}_f^\times$, and $\mathfrak{m}_1, \mathfrak{m}_2 \in \hat{\mathfrak{d}}^{-1}$, we define the global Kloosterman sum

$$S_{\omega_{\mathfrak{N}}}(\mathfrak{m}_1, \mathfrak{m}_2; \mathfrak{n}; \mathfrak{c}) = \sum_{\substack{s_1, s_2 \in \hat{\mathcal{O}}/\mathfrak{c}\hat{\mathcal{O}} \\ s_1 s_2 \equiv \mathfrak{n} \bmod \mathfrak{c}\hat{\mathcal{O}}}} \theta_f\Big(\frac{\mathfrak{m}_1 s_1 + \mathfrak{m}_2 s_2}{\mathfrak{c}}\Big)\omega_{\mathfrak{N}}(s_2)^{-1}, \tag{17}$$

We have the following relation between the global and local Kloosterman sums,

$$S_{\omega_{\mathfrak{N}}}(\mathfrak{m}_1, \mathfrak{m}_2; \mathfrak{n}; \mathfrak{c}) = \prod_{\nu<\infty} S_{\omega_{\mathfrak{N},\nu}}(\mathfrak{m}_{1\nu}, \mathfrak{m}_{2\nu}; \mathfrak{n}_\nu; \mathfrak{c}_\nu).$$

Note that the product on the RHS above is well-defined because $\mathfrak{c}_\nu \in \mathcal{O}_\nu^\times$ except for finitely many ν.

Let $K_f = \prod_{\nu<\infty} \mathrm{GL}_2(\mathcal{O}_\nu)$ be the standard maximal compact subgroup of $\mathrm{GL}_2(\mathbb{A}_f)$. Let

$$K_1(\mathfrak{N}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K_f : c \in \mathfrak{N}\hat{\mathcal{O}}, d \in 1 + \mathfrak{N}\hat{\mathcal{O}} \right\}, \tag{18}$$

and let $A_k(\mathfrak{N},\omega)$ be the space of Hilbert cusp forms with respect to $K_1(\mathfrak{N})$, of weight k and central character ω. We also define $K_0(\mathfrak{N}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in K_f : c \in \mathfrak{N}\hat{\mathcal{O}} \right\}$ and let

$$\psi(\mathfrak{N}) = [K_f : K_0(\mathfrak{N})] = \mathrm{Nm}(\mathfrak{N}) \prod_{\mathfrak{p}|\mathfrak{N}} \Big(1 + \frac{1}{\mathrm{Nm}(\mathfrak{p})}\Big).$$

For $\phi \in A_k(\mathfrak{N},\omega)$ and $m \in \mathfrak{d}_+^{-1}$, we now recall the definition of the mth Fourier coefficient of ϕ, denoted by W_m^ϕ, as in [KL08, §3.4]. Every character on $F\backslash\mathbb{A}$ is of the form $\theta_m(x) = \theta(-mx)$ for some $m \in F$. Consider the unipotent subgroup $\tilde{N} = \left\{ \begin{pmatrix} 1 & * \\ & 1 \end{pmatrix} \right\}$ of GL_2. As topological groups, $\tilde{N}(\mathbb{A}) \cong \mathbb{A}$, so characters of the two can be identified. Hence for $m \in F$, we identify θ_m with a character on $\tilde{N}(F)\backslash\tilde{N}(\mathbb{A})$ in the obvious way. For any $g \in \mathrm{GL}_2(\mathbb{A})$, the map $n \mapsto \phi(ng)$ is a continuous function on $\tilde{N}(F)\backslash\tilde{N}(\mathbb{A})$, with a Fourier expansion

$$\phi\Big(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g \Big) = \frac{1}{\sqrt{d_F}} \sum_{m \in F} W_m^\phi(g)\theta_m(x).$$

The coefficients are Whittaker functions defined by

$$W_m^\phi(g) = \int_{F\backslash\mathbb{A}} \phi\Big(\begin{pmatrix} 1 & x \\ & 1 \end{pmatrix} g \Big)\theta(mx)\,dx \tag{19}$$

where dx denotes the Lebesgue measure. For $y \in \mathbb{A}^\times$, we also denote

$$W_m^\phi(y) = W_m^\phi\Big(\begin{pmatrix} y & \\ & 1 \end{pmatrix} \Big),$$

ignoring the abuse of notation. We identify $\mathbb{A}_f^\times$ with $\{1_\infty\} \times \mathbb{A}_f^\times \subset \mathbb{A}^\times$, so that $y \in \mathbb{A}_f^\times$ can be identified as $\{1_\infty\} \times y$ in $\mathbb{A}^\times$.

We refer the reader to [KL08, Section 4.3] for the definition of the Hecke operators $T_\mathfrak{n}$ acting on $A_k(\mathfrak{N},\omega)$. We now recall the following result.

Lemma 4 ([KL08, Cor. 4.8])**.** *Let $\tilde{d} \in \mathbb{A}_f^\times$ such that $\tilde{d}\hat{\mathcal{O}} = \hat{\mathfrak{d}}$ and $m \in \mathfrak{d}_+^{-1}$. If $(m\mathfrak{d},\mathfrak{N}) = 1$, then for any $T_{m\mathfrak{d}}$-eigenfunction $\phi \in A_k(\mathfrak{N},\omega)$ with $W_1^\phi(1/\tilde{d}) = 1$ and $T_{m\mathfrak{d}}\phi = \lambda_{m\mathfrak{d}}^\phi\phi$, we have*

$$W_m^\phi(1) = \frac{e^{2\pi r}\prod_{j=1}^r \sigma_j(m)^{(k_j/2)-1}}{d_F e^{2\pi \mathrm{Tr}_\mathbb{Q}^F(m)}} \lambda_{m\mathfrak{d}}^\phi.$$

We now recall the statement of Petersson's trace formula in the Hilbert modular setting.

Theorem 5 ([KL08, Thm. 5.11])**.** *Let $\mathfrak{n}$ and $\mathfrak{N}$ be integral ideals with $(\mathfrak{n},\mathfrak{N}) = 1$. Let $k = (k_1,...,k_r)$ with all $k_j > 2$. Let $\mathcal{F}$ be an orthogonal basis for $A_k(\mathfrak{N},\omega)$ consisting of eigenfunctions for the Hecke operator $T_\mathfrak{n}$. Then for any $m_1, m_2 \in \mathfrak{d}_+^{-1}$, we have*

$$\begin{aligned}\frac{e^{2\pi\mathrm{Tr}_\mathbb{Q}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\left[\prod_{j=1}^r \frac{(k_j-2)!}{(4\pi\sqrt{\sigma_j(m_1m_2)})^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}} \frac{\lambda_\mathfrak{n}^\phi W_{m_1}^\phi(1)\overline{W_{m_2}^\phi(1)}}{\|\phi\|^2} &= \hat{T}(m_1,m_2,\mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_\mathfrak{N}(m_1/s)\omega_f(s)} \\ &+ \sum_{i=1}^t \sum_{\substack{u\in U \\ \eta_i u\in F^+}} \sum_{\substack{s\in \mathfrak{b}_i\mathfrak{N}/\pm \\ s\neq 0}} \Bigg\{ \omega_f(s\mathfrak{b}_i^{-1})S_{\omega_\mathfrak{N}}(m_1,m_2;\eta_i u\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1}) \\ &\times \frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)} \times \prod_{j=1}^r \frac{2\pi}{(\sqrt{-1})^{k_j}} J_{k_j-1}\left(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\right)\Bigg\}. \end{aligned} \tag{20}$$

- *where $\hat{T}(m_1,m_2,\mathfrak{n}) \in \{0,1\}$ is non zero if and only if there exists $s \in \hat{\mathfrak{d}}^{-1}$ such that $m_1m_2 \in s\hat{\mathcal{O}}$ and $m_1m_2\hat{\mathcal{O}} = s^2\hat{\mathfrak{n}}$,*
- *U is a set of representative for $\mathcal{O}^\times/\mathcal{O}^{\times 2}$,*
- *$\mathfrak{b}_i\hat{\mathcal{O}} = \hat{\mathfrak{b}}_i$ for integral ideal $\mathfrak{b}_i$ and $i = 1,\dots,t$, where $\mathfrak{b}_i$s are a set of representatives for the equation $[\mathfrak{b}]^2[\mathfrak{n}] = 1$ in the ideal class group,*
- *$\eta_i \in F$ generates the principal ideal $\mathfrak{b}_i^2\mathfrak{n}$,*
- *$\omega_\mathfrak{N} = \prod_{v|\mathfrak{N}}\omega_v\prod_{v\nmid\mathfrak{N}} 1$,*

Corollary 6. *Let $\tilde{d} \in \mathbb{A}_f^\times$ such that $\tilde{d}\hat{\mathcal{O}} = \hat{\mathfrak{d}}$ and $m_1, m_2 \in \mathfrak{d}_+^{-1}$ such that $(m_1\mathfrak{d}, \mathfrak{N}) = (m_2\mathfrak{d}, \mathfrak{N}) = 1$. Let $W_1^\phi(1/\tilde{d}) = 1$ for all $\phi \in \mathcal{F}$, where $\mathcal{F}$ is an orthogonal basis for $A_k(\mathfrak{N}, \omega)$. Then*

$$\begin{aligned}\frac{e^{4\pi r}}{\psi(\mathfrak{N})d_F^2\sqrt{\mathrm{Nm}(m_1m_2)}}\left[\prod_{j=1}^r\frac{(k_j-2)!}{(4\pi)^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{m_1\mathfrak{d}}^\phi\overline{\lambda_{m_2\mathfrak{d}}^\phi}}{\|\phi\|^2} &= \hat{T}(m_1,m_2,\mathcal{O})\frac{\sqrt{d_F}}{\omega_{\mathfrak{N}}(m_1/s)\omega_f(s)}\\ &+\sum_{i=1}^t\sum_{\substack{u\in U\\ \eta_iu\in F^+}}\sum_{\substack{s\in\mathfrak{b}_i\mathfrak{N}/\pm\\ s\neq 0}}\left\{\omega_f(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_iu\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_iu)}}{\mathrm{Nm}(s)}\right.\\ &\left.\times\prod_{j=1}^r\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\left(\frac{4\pi\sqrt{\sigma_j(\eta_ium_1m_2)}}{|\sigma_j(s)|}\right)\right\}.\end{aligned}$$

Proof. This corollary is proved by substituting the expressions for $W_{m_1}^\phi(1)$ and $W_{m_2}^\phi(1)$ obtained from Lemma 4 and taking $\mathfrak{n} = \mathcal{O}$ in Theorem 5. Using Lemma 4 we have

$$W_{m_1}^\phi(1) = \frac{e^{2\pi r}\prod_{j=1}^r\sigma_j(m_1)^{(k_j/2)-1}}{d_Fe^{2\pi\mathrm{Tr}_{\mathbb{Q}}^F(m_2)}}\lambda_{m_1\mathfrak{d}}^\phi$$

and

$$W_{m_2}^\phi(1) = \frac{e^{2\pi r}\prod_{j=1}^r\sigma_j(m_2)^{(k_j/2)-1}}{d_Fe^{2\pi\mathrm{Tr}_{\mathbb{Q}}^F(m_2)}}\lambda_{m_2\mathfrak{d}}^\phi.$$

Multiplying the first equation with the conjugate of the second one, we get

$$\begin{aligned}W_{m_1}^\phi(1)\overline{W_{m_2}^\phi(1)} &= \frac{e^{4\pi r}\prod_{j=1}^r\sigma_j(m_1m_2)^{(k_j/2)-1}}{d_F^2e^{2\pi\mathrm{Tr}_{\mathbb{Q}}^F(m_1+m_2)}}\lambda_{m_1\mathfrak{d}}^\phi\overline{\lambda_{m_2\mathfrak{d}}^\phi}\\ &= \frac{e^{4\pi r}\prod_{j=1}^r\sigma_j(m_1m_2)^{((k_j-1)/2)}}{d_F^2\sqrt{\prod_{j=1}^r\sigma_j(m_1m_2)}e^{2\pi\mathrm{Tr}_{\mathbb{Q}}^F(m_1+m_2)}}\lambda_{m_1\mathfrak{d}}^\phi\overline{\lambda_{m_2\mathfrak{d}}^\phi}.\end{aligned}$$

Thus

$$\frac{e^{2\pi\mathrm{Tr}_{\mathbb{Q}}^F(m_1+m_2)}}{\prod_{j=1}^r\sigma_j(m_1m_2)^{((k_j-1)/2)}}W_{m_1}^\phi(1)\overline{W_{m_2}^\phi(1)} = \frac{e^{4\pi r}}{d_F^2\sqrt{\mathrm{Nm}(m_1m_2)}}\lambda_{m_1\mathfrak{d}}^\phi\overline{\lambda_{m_2\mathfrak{d}}^\phi}.$$

In Theorem 5, taking $\mathfrak{n} = \mathcal{O}$ implies $\lambda_{\mathcal{O}}^\phi = 1$ and the proof is complete. □

Corollary 7. *Let $\mathfrak{p}$ be a prime ideal not dividing the level $\mathfrak{N}$. Let $k = (k_1, ..., k_r)$ with all $k_j > 2$. We also take an integer $\ell \geq 0$. Let $\mathcal{F}$ be an orthogonal basis for $A_k(\mathfrak{N}, \omega)$ consisting of eigenfunctions for the Hecke operator $T_{\mathfrak{p}^\ell}$. Then for any $m \in \mathfrak{d}_+^{-1}$, we have*

$$\begin{aligned}\frac{e^{4\pi\mathrm{Tr}_{\mathbb{Q}}^F(m)}}{\psi(\mathfrak{N})}\left[\prod_{j=1}^r\frac{(k_j-2)!}{(4\pi\sigma_j(m))^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{p}^\ell}^\phi|W_m^\phi(1)|^2}{\|\phi\|^2} &= \hat{T}(m,m,\mathfrak{p}^\ell)\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{p}^\ell)}}{\omega_{\mathfrak{N}}(m/s)\omega_f(s)}\\ &+\sum_{i=1}^t\sum_{\substack{u\in U\\ \eta_iu\in F^+}}\sum_{\substack{s\in\mathfrak{b}_i\mathfrak{N}/\pm\\ s\neq 0}}\left\{\omega_f(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m,m;\eta_iu\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_iu)}}{\mathrm{Nm}(s)}\right.\\ &\left.\times\prod_{j=1}^r\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\left(\frac{4\pi|\sigma_j(m)|\sqrt{\sigma_j(\eta_iu)}}{|\sigma_j(s)|}\right)\right\}.\end{aligned}$$

Proof. We obtain this corollary by taking $\mathfrak{n} = \mathfrak{p}^\ell$ and $m_1 = m_2 = m$ in Theorem 5. □

3. Estimating the error term in the trace formula

We prove Theorem 1 in this section. First, we estimate the triple sum appearing on the right-hand side of the trace formula given by Theorem 5. One of the main objects appearing in the sum is the J-Bessel function. Hence, in the next lemma, we recall bounds for the J-Bessel function of the first kind that will be useful.

Lemma 8. *We have the following estimates of the J-Bessel function.*

(i) If $a \geq 0$ and $0 < x \leq 1$, we have

$$1 \leq \frac{J_a(ax)}{x^a J_a(a)} \leq e^{a(1-x)}.$$

(ii) As $a \to \infty$, we have $0 < J_a(a) \ll \frac{1}{a^{\frac{1}{3}}}$.

(iii) If $|d| < 1$, then

$$\frac{1}{a^{\frac{1}{3}}} \ll J_a(a + da^{\frac{1}{3}}) \ll \frac{1}{a^{\frac{1}{3}}}.$$

(iv) For $x \in \mathbb{R}$ and $a > 0$, $|J_a(x)| \leq \min(ba^{-\frac{1}{3}}, c|x|^{-\frac{1}{3}})$ where $b \approx 0.674885$ and $c \approx 0.7857468704$.

(v) For $\frac{1}{2} \leq x < 1$, we have the following uniform bound

$$J_a(ax) \ll \frac{1}{(1-x^2)^{1/4} a^{1/2}}.$$

Proof. We refer to [JS20, §2.1.1] for all parts of the lemma, except (iv), for which we refer to [Lan00, §1]. Part (iii) is an essential ingredient for obtaining the lower bound in Theorem 2. The geometric origin of the transition behavior of the J-Bessel function given by (iii) is explained by Marshall in [JS20, §5]. □

We now give an outline of the proof of Theorem 1.

Outline of the proof of Theorem 1. In §3.1, we first fix i in equation (20), where i varies over a finite set. Then we take $u \in U$ satisfying $\eta_i u \in F^+$ to be fixed till §3.3. This helps us to focus on a single lattice at a time and estimate the right-hand side of equation (20). In §3.2, we fix an orthant of the lattice so that we can apply J-Bessel function estimates carefully. This is necessary as we do not use a uniform bound for a fixed lattice. According to the usability of Lemma 8, we apply bounds for a fixed orthant in two steps. In §3.3, firstly we vary over all orthants (earlier fixed in §3.2) for a fixed lattice, then we vary over all possible lattices along with $u \in U$ satisfying $\eta_i u \in F^+$ to complete the proof.

Proof of Theorem 1. Let m_1 and m_2 satisfy equation (14). We fix some notations before we proceed further in the proof. Let δ_i and A_i be as in equations (12) and (13). Recall that

$$\gamma_j = \max\left\{\sqrt{\sigma_j(\eta_i u)} \,:\, i = 1, \dots, t, u \in U, \eta_i u \in F^+\right\}.$$

Further, let

$$\beta_j = \min\left\{\sqrt{\sigma_j(\eta_i u)} \,:\, i = 1, \dots, t, u \in U, \eta_i u \in F^+\right\} \qquad \text{and} \qquad \epsilon_j = \frac{\gamma_j}{\beta_j}$$

for all $j = 1, \dots, r$.

3.1. **Fixing a lattice.** Let us fix i between 1 and t in the triple sum given in equation (20). Note that η_i is now fixed, as i is fixed. We now fix $u \in U$ satisfying $\eta_i u \in F^+$ until §3.3. We break the last of the triple sum into two parts, the first part is a summation over $s \in A_i$ and and the second over $s \notin A_i$. By taking $A_i' = \{s \in \mathfrak{b}_i\mathfrak{N}/\pm \,:\, s \neq 0, s \notin A_i\}$, we break down the summation

$$\sum_{\substack{s \in \mathfrak{b}_i\mathfrak{N}/\pm, \\ s \neq 0}} \omega_{\mathrm{f}}(s\mathfrak{b}_i^{-1}) S_{\omega_{\mathfrak{N}}}(m_1, m_2; \eta_i u \mathfrak{b}_i^{-2}; s\mathfrak{b}_i^{-1}) \frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)} \prod_{j=1}^{r} \frac{2\pi}{(\sqrt{-1})^{k_j}} J_{k_j - 1}\left(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\right)$$

into summations over

$$\sum_{s \in A_i} \quad \text{and} \quad \sum_{s \in A_i'}.$$

As A_i is finite, the sum $\sum_{s \in A_i}$ is finite and is going to constitute the main term. So we focus on estimating the sum when the sum is taken over $s \in A_i'$. We show that as $k_0 \to \infty$, the sum

$$\sum_{s \in A_i'} \omega_{\mathrm{f}}(s\mathfrak{b}_i^{-1}) S_{\omega_{\mathfrak{N}}}(m_1, m_2; \eta_i u \mathfrak{b}_i^{-2}; s\mathfrak{b}_i^{-1}) \frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)} \prod_{j=1}^{r} \frac{2\pi}{(\sqrt{-1})^{k_j}} J_{k_j - 1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big) \tag{21}$$

$$= \mathrm{O}\left(\sum_{j=1}^{r}\left((k_j - 1)^{-\frac{1}{2}} \prod_{j'=1, j' \neq j} (k_{j'} - 1)^{-\frac{1}{3}}\right)\right).$$

The above asymptotic is achieved in equation (44).

Using Lemma 6.1 from [KL08], we get

$$\Big|S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_i u\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\Big| \leq \mathrm{Nm}(\eta_i u\mathfrak{b}_i^{-2})\mathrm{Nm}(s\mathfrak{b}_i^{-1}),$$

which implies

$$\left|\omega_{\mathrm{f}}(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_i u\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)}\right| \leq \mathrm{Nm}(\eta_i u)^{\frac{3}{2}}\mathrm{Nm}(\mathfrak{b}_i^{-3}),$$

and consequently

$$\left|\sum_{s\in A_i'}\omega_{\mathrm{f}}(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_i u\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)}\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right|$$

$$\leq \sum_{s\in A_i'}\mathrm{Nm}(\eta_i u)^{\frac{3}{2}}\mathrm{Nm}(\mathfrak{b}_i^{-3})\prod_{j=1}^{r}2\pi\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right|. \tag{22}$$

Therefore, our goal now is to find bounds for

$$\sum_{s\in A_i'}\prod_{j=1}^{r}\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right|.$$

Since $s\in A_i'$, there exists j_0 (j_0 depends upon s) such that $|\sigma_{j_0}(s)|>2\tilde{\delta}_i$. Due to the hypothesis

$$\frac{2\pi\gamma_{j_0}\sqrt{\sigma_{j_0}(m_1m_2)}}{\tilde{\delta}_i}\in\Big((k_{j_0}-1)-(k_{j_0}-1)^{\frac{1}{3}},(k_{j_0}-1)\Big),$$

we have the inequality

$$\frac{8}{9}<\left(1-\frac{1}{(k_{j_0}-1)^{\frac{2}{3}}}\right)<\left|\frac{2\pi\gamma_{j_0}\sqrt{\sigma_{j_0}(m_1m_2)}}{(k_{j_0}-1)\tilde{\delta}_i}\right|<1, \tag{23}$$

whenever $k_{j_0}>28$. Applying part (i) of Lemma 8, we have

$$\left|J_{k_{j_0}-1}\Big(\frac{4\pi\sqrt{\sigma_{j_0}(\eta_i u m_1 m_2)}}{|\sigma_{j_0}(s)|}\Big)\right|=\left|J_{k_{j_0}-1}\left((k_{j_0}-1)\frac{4\pi\sqrt{\sigma_{j_0}(\eta_i u m_1 m_2)}}{(k_{j_0}-1)|\sigma_{j_0}(s)|}\right)\right|\leq e^{a(1-x)}x^a J_a(a),$$

by taking $a=k_{j_0}-1$ and $x=\frac{4\pi\sqrt{\sigma_{j_0}(\eta_i u m_1 m_2)}}{(k_{j_0}-1)|\sigma_{j_0}(s)|}$. The lemma is applicable since

$$x<\frac{2\pi\gamma_{j_0}\sqrt{\sigma_{j_0}(m_1m_2)}}{(k_{j_0}-1)\tilde{\delta}_i}<1$$

by equation (23). Using part (ii) of Lemma 8, we obtain

$$J_a(a)\ll\frac{1}{a^{\frac{1}{3}}}=\frac{1}{(k_{j_0}-1)^{\frac{1}{3}}}. \tag{24}$$

Moreover, equation (23) implies that $\frac{16\tilde{\delta}_i}{(9\epsilon_{j_0})|\sigma_{j_0}(s)|}<x<\frac{2\tilde{\delta}_i}{|\sigma_{j_0}(s)|}$. Therefore, we obtain

$$e^{a(1-x)}x^a=e^{a(1-x+\log x)}<e^{a\left(1-\frac{16\tilde{\delta}_i}{(9\epsilon_{j_0})|\sigma_{j_0}(s)|}+\log\left(\frac{2\tilde{\delta}_i}{|\sigma_{j_0}(s)|}\right)\right)}. \tag{25}$$

We now need a device to keep track of the place where $|\sigma_j(s)|>2\tilde{\delta}_i$. The following consideration helps us to achieve this. Let $M>1$ be an absolute constant, suitably chosen later. For each j, the quantity $\sigma_j(s)$ falls into one of the following three categories:

- $|\sigma_j(s)|\leq 2\tilde{\delta}_i$,
- $2\tilde{\delta}_i<|\sigma_j(s)|\leq(M+1)\tilde{\delta}_i$, or
- $|\sigma_j(s)|>(M+1)\tilde{\delta}_i$.

We then consider an r-tuple $(a_1,\ldots,a_r)$ with each $a_j\in\{0,1,2\}$ defined as follows:

$$a_j=\begin{cases}0, & \text{if } |\sigma_j(s)|\leq 2\tilde{\delta}_i,\\ 1, & \text{if } 2\tilde{\delta}_i<|\sigma_j(s)|\leq(M+1)\tilde{\delta}_i,\\ 2, & \text{if } |\sigma_j(s)|>(M+1)\tilde{\delta}_i.\end{cases} \tag{26}$$

For each $\ell\in\{0,1,2\}$, let $\mathfrak{I}_\ell=\{j\mid a_j=\ell\}$. Clearly, $\{1,\ldots,r\}=\mathfrak{I}_0\sqcup\mathfrak{I}_1\sqcup\mathfrak{I}_2$.

For each r-tuple $(a_1, \dots, a_r)$ as above, we set

$$h = \sum_{j=1}^{r} a_j 3^{j-1}. \tag{27}$$

For each $s \neq 0$, we have the corresponding a_j depending on the value of $|\sigma_j(s)|$. Therefore, we have a corresponding $h(s)$. Note that $s \in A_i$ if and only if $h(s) = 0$. We now partition the set A_i' according to the values of $h(s)$. Let $A_{i,h}'$ denote the subset of A_i' consisting of all s such that $h(s) = h$. Finally, we have

$$A_i' = \cup_{h=1}^{3^r-1} A_{i,h}'.$$

Hence, we have

$$\sum_{s \in A_i'} = \sum_{h=1}^{3^r-1} \sum_{s \in A_{i,h}'}. \tag{28}$$

For a fixed h (recall that i is already fixed), we now find bounds for the expression

$$\sum_{s \in A_{i,h}'} \prod_{j=1}^{r} \left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right|. \tag{29}$$

3.1.1. *Bounds for* $\mathfrak{I}_0$ *and* $\mathfrak{I}_2$. For $j \in \mathfrak{I}_0$, we use the uniform bound given by Lemma 8(iv). We have

$$\left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \leq \min\left((k_j-1)^{-\frac{1}{3}}, \Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)^{-\frac{1}{3}}\right)$$

$$\ll \min\left((k_j-1)^{-\frac{1}{3}}, \Big(\sqrt{\sigma_j(\eta_i u m_1 m_2)}\Big)^{-\frac{1}{3}}\right)$$

as $|\sigma_j(s)| \leq 2\tilde{\delta}_i$. By equation (23), $\sqrt{\sigma_j(\eta_i u m_1 m_2)} \geq \frac{4\sqrt{\sigma_j(\eta_i u)}(k_j-1)\tilde{\delta}_i}{9\pi\gamma_j}$, which after using in the above upper bound yields

$$\left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \ll_{F,\mathfrak{N},\mathfrak{n}} \min\Big((k_j-1)^{-\frac{1}{3}}, (k_j-1)^{-\frac{1}{3}}\Big) = (k_j-1)^{-\frac{1}{3}}. \tag{30}$$

Now for $j \in \mathfrak{I}_2$ equation (24) and (25) implies

$$\left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \leq (k_j-1)^{-\frac{1}{3}} e^{(k_j-1)\left(1-\frac{16\tilde{\delta}_i}{(9\epsilon_j)|\sigma_j(s)|}+\log\left(\frac{2\tilde{\delta}_i}{|\sigma_j(s)|}\right)\right)}. \tag{31}$$

3.2. **Estimates of the J-Bessel functions.** We consider the sum given in equation (29),

$$\sum_{s \in A_{i,h}'} \prod_{j=1}^{r} \left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right|.$$

We can express this as

$$\sum_{s \in A_{i,h}'} \prod_{j \in \mathfrak{I}_0} \prod_{j \in \mathfrak{I}_1} \prod_{j \in \mathfrak{I}_2}$$

by breaking the inner product into products over the three sets $\mathfrak{I}_0$, $\mathfrak{I}_1$ and $\mathfrak{I}_2$. It is important to note that the partition of $\{1, \dots, r\}$ into $\mathfrak{I}_0$, $\mathfrak{I}_1$ and $\mathfrak{I}_2$ does not depend on s, as s varies in the lattice $A_{i,h}'$. This is essential to apply Lemma 8 smoothly in a phased manner.

We now apply estimates for the J-Bessel function. We break this into two cases. In the first case, the set $\mathfrak{I}_1 \neq \emptyset$ and we address this case in §3.2.1 below. In the second case, the set $\mathfrak{I}_1 = \emptyset$, and this case is addressed in §3.2.2 below.

3.2.1. *Case:* $\mathfrak{I}_1 \neq \emptyset$. For any $j \in \mathfrak{I}_1$, let

$$\tilde{x}_j(s) = \frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{(k_j-1)|\sigma_j(s)|}. \tag{32}$$

Note that $0 < \tilde{x}_j(s) < 1$. We will be applying different bounds depending on whether $0 < \tilde{x}_j(s) < 1/3$, $1/3 \leq \tilde{x}_j(s) < 1/2$ or $1/2 \leq \tilde{x}_j(s) < 1$.

We temporarily drop s from the notation and use $\tilde{x}_j$ for simplicity. When $1/2 \leq \tilde{x}_j < 1$, we use the uniform bound given by Lemma 8, part (v). This gives

$$J_{k_j-1}((k_j-1)\tilde{x}_j) \ll \frac{1}{(1-\tilde{x}_j^2)^{\frac{1}{4}}(k_j-1)^{\frac{1}{2}}} \ll_{\mathfrak{n},\mathfrak{N},F} (k_j-1)^{-\frac{1}{2}}. \tag{33}$$

The last step is justified due to the fact $\tilde{x}_j(s)$ cannot be arbitrarily close to 1 as s varies over a discrete subset of $\mathbb{R}^r$ and $2\tilde{\delta}_i < |\sigma_j(s)| \leq (M+1)\tilde{\delta}_i$.

When $1/3 \leq \tilde{x}_j < 1/2$, we use Lemma 8, parts (i) and (ii) to obtain

$$J_{k_j-1}((k_j-1)\tilde{x}_j) \ll_{\mathfrak{n},\mathfrak{N},F} e^{(k_j-1)(1-\tilde{x}_j+\log\tilde{x}_j)} \cdot \frac{1}{(k_j-1)^{\frac{1}{3}}}$$

$$\ll_{\mathfrak{n},\mathfrak{N},F} e^{(k_j-1)(1-\frac{1}{3}+\log\frac{1}{2})} \cdot \frac{1}{(k_j-1)^{\frac{1}{3}}}. \tag{34}$$

Finally, for $0 < \tilde{x}_j < 1/3$, we obtain

$$J_{k_j-1}((k_j-1)\tilde{x}_j) \ll_{\mathfrak{n},\mathfrak{N},F} e^{(k_j-1)(1+\log\frac{1}{3})} \cdot \frac{1}{(k_j-1)^{\frac{1}{3}}}. \tag{35}$$

By considering equation (33), (34) and (35), for all indices $j \in \mathfrak{I}_1$, we obtain

$$J_{k_j-1}\left(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\right) \ll_{\mathfrak{n},\mathfrak{N},F} (k_j-1)^{-\frac{1}{2}}. \tag{36}$$

Let $\tau_{i,1}, ..., \tau_{i,r}$ denote the lengths of the sides of the fundamental parallelepiped of the lattice $\sigma(b_i\mathfrak{N})$. Let $\tilde{\tau}_i = \min(\tau_{i,1}, \ldots, \tau_{i,r})$ and $\tilde{\epsilon}_i = \left(\frac{\tilde{\tau}_i}{2}\right)^r$. By the choice of $\tilde{\epsilon}_i$, we note that a cube of volume $\tilde{\epsilon}_i$ can contain at most one lattice point of $\sigma(\mathfrak{b}_i\mathfrak{N})$, and hence also at most one lattice point of $\sigma(A'_{i,h})$. Using the bounds in 36, we obtain

Note that since the partition $\{1, \ldots, r\} = \mathfrak{I}_0 \cup \mathfrak{I}_1 \cup \mathfrak{I}_2$ is the same for any $s \in A'_{i,h}$, we can interchange the summation and the products to get

$$\sum_{s\in A'_{i,h}} \prod_{j\in\mathfrak{I}_0} \prod_{j\in\mathfrak{I}_1} \prod_{j\in\mathfrak{I}_2} \left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| = \prod_{j\in\mathfrak{I}_0} \prod_{j\in\mathfrak{I}_1} \prod_{j\in\mathfrak{I}_2} \sum_{s\in A'_{i,h}} \left| J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right|$$

Now, if $j \in \mathfrak{I}_0$, then $|\sigma_j(s)| \in (0, 2\tilde{\delta}_i]$. We write this interval as a union of smaller intervals of length $\tilde{\tau}_i/2$. That is,

$$(0, 2\tilde{\delta}_i] = \cup_{\tilde{m}_j} \left(\frac{(\tilde{m}_j-1)\tilde{\tau}_i}{2}, \frac{\tilde{m}_j\tilde{\tau}_i}{2}\right].$$

Note that there are only finitely many possible $\tilde{m}_j$ as the original interval $(0, 2\tilde{\delta}_i]$ is of finite length. Similarly, if $j \in \mathfrak{I}_1$ or $j \in \mathfrak{I}_2$, then $|\sigma_j(s)| \in (2\tilde{\delta}_i, (M+1)\tilde{\delta}_i]$ or $|\sigma_j(s)| \in ((M+1)\tilde{\delta}_i, \infty)$, respectively. In each of these cases, we again break up the intervals into smaller ones of length $\tilde{\tau}_i/2$. That is, we write

$$(2\tilde{\delta}_i, (M+1)\tilde{\delta}_i] = \cup_{\tilde{m}_j} \left(2\tilde{\delta}_i + \frac{(\tilde{m}_j-1)\tilde{\tau}_i}{2}, 2\tilde{\delta}_i + \frac{\tilde{m}_j\tilde{\tau}_i}{2}\right]$$

and

$$((M+1)\tilde{\delta}_i, \infty) = \cup_{\tilde{m}_j} \left((M+1)\tilde{\delta}_i + \frac{(\tilde{m}_j-1)\tilde{\tau}_i}{2}, (M+1)\tilde{\delta}_i + \frac{\tilde{m}_j\tilde{\tau}_i}{2}\right].$$

The number of $\tilde{m}_j$ in the first two cases is finite and infinite in the last case.

Now, we take $\tilde{m} = (\tilde{m}_j)_{j=1}^r$ and consider the corresponding interval $I_{\tilde{m}_j} = \left(* + \frac{(\tilde{m}_j-1)\tilde{\tau}_i}{2}, * + \frac{\tilde{m}_j\tilde{\tau}_i}{2}\right]$ as above. Then, $I_{\tilde{m}} = \prod_{j=1}^r I_{\tilde{m}_j} \subset \mathbb{R}^r$ is hypercube of volume $\left(\frac{\tau_i}{2}\right)^r = \tilde{\epsilon}_i$ and hence contains at most one lattice point of $\sigma(A'_{i,h})$.

$$\sum_{s\in A'_{i,h}}\prod_{j\in\mathfrak{I}_0}\prod_{j\in\mathfrak{I}_1}\prod_{j\in\mathfrak{I}_2}\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \ll \prod_{j\in\mathfrak{I}_0}\prod_{j\in\mathfrak{I}_1}\prod_{j\in\mathfrak{I}_2}\sum_{\substack{\tilde{m},\\ \sigma(s)\in I_{\tilde{m}}}}\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \tag{37}$$

By using equations (30) for $j\in\mathfrak{I}_0$, (31) for $j\in\mathfrak{I}_2$ and (36) for $j\in\mathfrak{I}_1$ the RHS of (37) is bounded by

$$\ll \prod_{j\in\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}\prod_{j\in\mathfrak{I}_1}(k_j-1)^{-\frac{1}{2}}\prod_{j\in\mathfrak{I}_2}(k_j-1)^{-\frac{1}{3}}\sum_{\substack{\tilde{m}_j,\\ \sigma_j(s)\in I_{\tilde{m}_j}}} e^{(k_j-1)\left(1-\frac{16\delta_i}{9\epsilon_j|\sigma_j(s)|}+\log\left(\frac{2\delta_i}{|\sigma_j(s)|}\right)\right)} \tag{38}$$

Taking $y_j=y_j(s)=\frac{|\sigma_j(s)|}{\delta_j}$, the sum above becomes

$$= \prod_{j\in\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}\prod_{j\in\mathfrak{I}_1}(k_j-1)^{-\frac{1}{2}}\prod_{j\in\mathfrak{I}_2}(k_j-1)^{-\frac{1}{3}}\sum_{\substack{\tilde{m}_j,\\ \sigma_j(s)\in I_{\tilde{m}_j}}} e^{(k_j-1)\left(1-\frac{16}{9\epsilon_j y_j}+\log\left(\frac{2}{y_j}\right)\right)} \tag{39}$$

Note that each of the intervals $I_{\tilde{m}_j}$ has length $|I_{\tilde{m}_j}|=\epsilon^{1/r}/\delta_i$. Therefore, the last equation becomes

$$= \frac{\delta_i^r}{\epsilon_i}\prod_{j\in\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}\prod_{j\in\mathfrak{I}_1}(k_j-1)^{-\frac{1}{2}}\prod_{j\in\mathfrak{I}_2}(k_j-1)^{-\frac{1}{3}}\sum_{\substack{\tilde{m}_j,\\ \sigma_j(s)\in I_{\tilde{m}_j}}} e^{(k_j-1)\left(1-\frac{16}{9\epsilon_j y_j}+\log\left(\frac{2}{y_j}\right)\right)}|I_{\tilde{m}_j}| \tag{40}$$

Note that for $j\in\mathfrak{I}_0,\mathfrak{I}_1$ and $\mathfrak{I}_2$, the variables y_j belong to the intervals $(0,2]$, $(2,M+1]$ and $(M+1,\infty)$ respectively. We denote these intervals by I_j. For $j\in\mathfrak{I}_0$ and $\mathfrak{I}_1$, we also denote by $\tilde{I}_j$ the slightly larger intervals $(a_j,b_j+\frac{\epsilon_i^{\frac{1}{r}}}{\delta_i}]$, where $I_j=(a_j,b_j]$. For $j\in\mathfrak{I}_2$, we set $\tilde{I}_j=I_j$. Since

$$e^{(k_j-1)\left(-\frac{16}{9\epsilon_j y_j}+\log\left(\frac{2}{y_j}\right)\right)}$$

is a monotonically decreasing function of y_j for $y_j>2$, using the integral test, the above sum is bounded by the multiple integral

$$\leq \frac{\delta_i^r}{\epsilon_i}\left(\prod_{j\in\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}\prod_{j\in\mathfrak{I}_1}(k_j-1)^{-\frac{1}{2}}\int_{\tilde{I}_j}\cdots\int\prod dy_j\right)$$
$$\left(\prod_{j\in\mathfrak{I}_2}(k_j-1)^{-\frac{1}{3}}\int_{\tilde{I}_j}\cdots\int e^{(k_j-1)\left(1-\frac{16}{9\epsilon_j y_j}+\log\left(\frac{2}{y_j}\right)\right)}\prod dy_j\right).$$

By absorbing some absolute constants, the above is

$$\ll \left(\prod_{j\in\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}\prod_{j\in\mathfrak{I}_1}(k_j-1)^{-\frac{1}{2}}\right)\left(\prod_{j\in\mathfrak{I}_2}(k_j-1)^{-\frac{1}{3}}\int_M^\infty e^{(k_j-1)\left(1-\frac{16}{9\epsilon_j y_j}+\log\left(\frac{2}{y_j}\right)\right)}dy_j\right). \tag{41}$$

Now, for a fixed $j\in\mathfrak{I}_2$, consider the integral

$$\int_M^\infty e^{(k_j-1)\left(-\frac{16}{9\epsilon_j y_j}+\log\left(\frac{2}{y_j}\right)\right)}dy_j.$$

By substituting $x_j = \frac{16}{9\epsilon_j y_j}$ and $dx_j = -\frac{16dy_j}{9\epsilon_j y_j^2}$, the integral becomes

$$\begin{aligned}
&\left(\frac{9\epsilon_j}{4}\right)\int_0^{\frac{16}{9\epsilon_j M}} \left(e^{-(k_j-1)x_j}\right)\left(\frac{9\epsilon_j x_j}{8}\right)^{k_j-3} dx_j \\
&\leq \left(\frac{9\epsilon_j}{4}\right)\int_0^{\frac{16}{9\epsilon_j M}} e^{-(k_j-1)x_j}\left(\frac{2}{M}\right)^{k_j-3} dx_j \\
&= \left(\frac{9\epsilon_j}{4}\right)\left(\frac{2}{M}\right)^{k_j-3}\int_0^{\frac{16}{9\epsilon_j M}} e^{-(k_j-1)x_j}\, dx_j \\
&= \left(\frac{9\epsilon_j}{4}\right)\left(\frac{2}{M}\right)^{k_j-3}\left(\frac{e^{-(k_j-1)\frac{16}{9\epsilon_j M}}}{-(k_j-1)} + \frac{1}{k_j-1}\right).
\end{aligned}$$

Finally, putting together equations (37) and (41), we obtain

$$\sum_{s\in A'_{i,h}}\prod_{j=1}^{r}\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \ll \prod_{j\in\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}\prod_{j\in\mathfrak{I}_1}(k_j-1)^{-\frac{1}{2}}\prod_{j\in\mathfrak{I}_2}\left(\frac{2e}{M}\right)^{(k_j-3)}(k_j-1)^{-\frac{4}{3}}, \quad (42)$$

in the case $\mathfrak{I}_1 \neq \emptyset$.

3.2.2. *Case:* $\mathfrak{I}_1 = \emptyset$. In this case, we use the bounds in §3.1.1 and the arguments above to obtain

$$\sum_{s\in A'_{i,h}}\prod_{j=1}^{r}\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \ll \prod_{\mathfrak{I}_2}\left(\frac{2e}{M}\right)^{(k_j-3)}(k_j-1)^{-\frac{4}{3}}\prod_{\mathfrak{I}_0}(k_j-1)^{-\frac{1}{3}}. \quad (43)$$

3.3. **Putting everything together.** We now use (28) and the bounds obtained above to bound the sum over $s \in A'_i$.

For any $1 \leq \rho \leq r$, suppose that $h \in [3^{\rho-1}, 3^\rho - 1]$. The tuple corresponding to $h = 3^{\rho-1}$ is $(\delta_{j\rho})$. In this case $\mathfrak{I}_1 = \{\rho\}$ and $\mathfrak{I}_0 = \{1, \ldots, r\} \setminus \{\rho\}$. The contribution of this h to the sum is bounded above by

$$(k_\rho - 1)^{-\frac{1}{2}}\prod_{j\neq\rho}(k_j-1)^{-\frac{1}{3}}.$$

Now, consider $h \in (3^{\rho-1}, 3^\rho - 1]$, and its corresponding r-tuple $(a_1, \ldots, a_r)$. We have $a_j = 0$ for all $j > \rho$. For $j \leq \rho$, either $a_j = 0$ for all $j < \rho$ (and in that case $a_\rho = 2$), or $a_j = 1$ or 2 for some $j < \rho$. In the former case, the contribution of this h to the sum is bounded above by

$$\left(\frac{2e}{M}\right)^{(k_\rho-3)}(k_\rho-1)^{-\frac{4}{3}}\prod_{j\neq\rho}(k_j-1)^{-\frac{1}{3}}.$$

In the latter case, contribution of this h to the sum is bounded above by

$$\mathrm{o}\left((k_\rho-1)^{-\frac{1}{2}}\prod_{'\neq\rho}(k_j-1)^{-\frac{1}{3}}\right).$$

Therefore, for any $h \in [3^{\rho-1}, 3^\rho - 1]$, the contribution of h to the sum is bounded above by

$$(k_\rho - 1)^{-\frac{1}{2}}\prod_{j\neq\rho}(k_j-1)^{-\frac{1}{3}},$$

and the dominant terms are when $h = 3^{\rho-1}$. Varying h from 1 to $3^r - 1$ and collecting dominant terms, we obtain

$$\sum_{s\in A'_i}\prod_{j=1}^{r}\left|J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)\right| \ll \sum_{\rho=1}^{r}\left((k_\rho-1)^{-\frac{1}{2}}\prod_{j\neq\rho}(k_j-1)^{-\frac{1}{3}}\right). \quad (44)$$

By combining equations (22) and (44), for fixed i and $u \in U$ with $\eta_i u \in F^+$, we have

$$\begin{aligned}
&\sum_{s\in\mathfrak{b}_i\mathfrak{N}/\pm, s\neq 0}\omega_{\mathrm{f}}(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_i u\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)}\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big) \\
&= \sum_{s\in A_i}\omega_{\mathrm{f}}(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_i u\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_i u)}}{\mathrm{Nm}(s)}\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_i u m_1 m_2)}}{|\sigma_j(s)|}\Big)
\end{aligned}$$

$$+\mathrm{O}\left(\sum_{\rho=1}^{r}\left((k_\rho-1)^{-\frac{1}{2}}\prod_{j\neq\rho}(k_j-1)^{-\frac{1}{3}}\right)\right).$$

By varying i and $u\in U$ with $\eta_i u\in F^+$, we obtain the following refinement of (20):

$$\frac{e^{2\pi tr_{\mathbb{Q}}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\left[\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi\sqrt{\sigma_j(m_1m_2)})^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{n}}^{\phi}W_{m_1}^{\phi}(1)\overline{W_{m_2}^{\phi}(1)}}{\|\phi\|^2}=\hat{T}(m_1,m_2,\mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_{\mathfrak{N}}(m_1/s)\omega_{\mathfrak{f}}(s)}$$

$$+\sum_{i=1}^{t}\sum_{u\in U,\eta_iu\in F^+}\sum_{s\in A_i}\left\{\omega_{\mathfrak{f}}(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_iu\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\right.$$

$$\left.\frac{\sqrt{\mathrm{Nm}(\eta_iu)}}{\mathrm{Nm}(s)}\times\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_ium_1m_2)}}{|\sigma_j(s)|}\Big)\right\}+\mathrm{O}_{A_i,u,\eta_i}\left(\sum_{j=1}^{r}\left((k_j-1)^{-\frac{1}{2}}\prod_{j'\neq j}(k_{j'}-1)^{-\frac{1}{3}}\right)\right).$$

This completes the proof of Theorem 1. □

4. Main term in special cases: Proof of Theorem 2

This section considers the main term in Theorem 1, namely, the sum

$$\hat{T}(m_1,m_2,\mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_{\mathfrak{N}}(m_1/s)\omega_{\mathfrak{f}}(s)}+\sum_{i=1}^{t}\sum_{u\in U,\eta_iu\in F^+}\sum_{s\in A_i}\left\{\omega_{\mathfrak{f}}(s\mathfrak{b}_i^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_iu\mathfrak{b}_i^{-2};s\mathfrak{b}_i^{-1})\right.$$

$$\left.\frac{\sqrt{\mathrm{Nm}(\eta_iu)}}{\mathrm{Nm}(s)}\times\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_ium_1m_2)}}{|\sigma_j(s)|}\Big)\right\},$$

in special cases. We wish to obtain a lower bound for this, which involves a triple sum. We are able to do this in special cases when the triple sum simplifies to a single term.

We now assume that the field F has odd narrow class number. Note that, for any abelian group A, the quotient group A/A^2 is 2-torsion. If the quotient is a finite group, then its cardinality is $|A/A^2|=2^a$ and we denote this exponent as $\dim_2(A):=a$. If the group is infinite, then we denote $\dim_2(A)=\infty$. Recall that we have $U=\mathcal{O}^\times/\mathcal{O}^{\times 2}$ and we let $U^+=U\cap F^+$. We also denote the narrow class group of F by C^+. Recall that the narrow class group is the set of equivalence classes of fractional ideals up to totally positive principal ideals.

Recall that t denotes the number of solutions to the equation $[\mathfrak{b}]^2[\mathfrak{n}]=1$ in the ideal class group.

Lemma 9 ([EMP86, Prop. 2.4])**.** *If F has odd narrow class number, then $t=1$ and $|U^+|=1$.*

Proof. Since the class number is a divisor of the narrow class number, the class number is odd. As in [KL08, Example 5.16], the equation $[\mathfrak{b}]^2[\mathfrak{n}]=1$ has a unique solution, which implies $t=1$. By [EMP86, Proposition 2.4], we have $\dim_2(U^+)\leq\dim_2(C^+)$. Since, the latter is 0, this implies that $|U^+|=1$. □

Lemma 9 implies that we may take $\eta_1=1$, and $|\{u\in U:\eta_iu\in F^+ \text{ for some } i=1,\dots,t\}|=1$. Note that the set A_1 still need not be a singleton set. In the next few lemmas, we illustrate some cases in which $|A_1|=1$ and in those cases $\sum_{s\in A_1}$ is a single term.

Lemma 10. *Suppose the $\inf\{\|\sigma(s)\|:s\in\mathfrak{b}_1\mathfrak{N}/\pm,s\neq 0\}=\delta_1$ is attained for some s_0 such that $\sigma(s_0)=(a,a,\dots,a)=\left(\frac{\delta_1}{\sqrt{r}},\frac{\delta_1}{\sqrt{r}},\dots,\frac{\delta_1}{\sqrt{r}}\right)$. Then $A_1=\left\{\sigma(s_0)\right\}$.*

Proof. Let $s\in A_1$. We have $|\sigma_j(s)|\leq\frac{\delta_1}{\sqrt{r}}$ for all j. Hence $A_1\subset S_{\delta_1}$, where

$$S_{\delta_1}=\{s\in\mathfrak{b}_1\mathfrak{N}/\pm:\sigma_1^2(s)+\cdots+\sigma_r^2(s)\leq\delta_1^2\}.$$

Let $s'\in A_1\cap S_{\delta_1}$. By the choice of δ_1, we have $s'\in\partial A_1\cap S_{\delta_1}$ and hence $|\sigma_j(s')|=\frac{\delta_1}{\sqrt{r}}$ for all $j=1,...,r$. Therefore s' is equal to $\left((-1)^{m_1}\frac{\delta_1}{\sqrt{r}},(-1)^{m_2}\frac{\delta_1}{\sqrt{r}},\dots,(-1)^{m_r}\frac{\delta_1}{\sqrt{r}}\right)$ with $m_j=0,1$ for $j=1,\dots,r$. If $m_j=0$ or $m_j=1$ for all j, then $s'=s$. In any other case, we have $m_{j_0}=0$ for some j_0, so that $\sigma_{j_0}(s)=\sigma_{j_0}(s')$. However, σ_{j_0} injective implies $s'=s_0=\frac{\delta_1}{\sqrt{r}}$. □

Lemma 11. *Let $\mathfrak{b}_1\mathfrak{N}=\mathcal{O}$. Then $\delta_1=\sqrt{r}$ and $A_1=\{\sigma(1)\}$.*

Proof. We wish to minimize $\inf\{\|\sigma(s)\| : s \in \mathcal{O}/\pm, s \neq 0\}$. This is equivalent to minimize $\|\sigma(s)\|^2 = \sigma_1^2(s) + \cdots + \sigma_r^2(s)$. For a given s, let

$$\sigma(s') = ((-1)^{m_1}\sigma_1(s), (-1)^{m_2}\sigma_2(s), \ldots, (-1)^{m_r}\sigma_r(s)).$$

Then, $\|\sigma(s')\| = \|\sigma(s)\|$. Thus, without loss of generality, we can consider minimizing $\|\sigma(s)\|^2$ on the set $\{s : s \in \mathcal{O}/\pm, s \neq 0, \sigma_j(s) \geq 0 \text{ for } j = 1, \ldots, r\}$. Using the Cauchy-Schwartz inequality, we have

$$\sqrt{(1^2 + \ldots + 1^2)(\sigma_1^2(s) + \ldots + \sigma_r^2(s))} \geq \sigma_1(s) + \ldots + \sigma_r(s).$$

Using the AM-GM inequality,

$$\frac{\sigma_1(s) + \ldots + \sigma_r(s)}{r} \geq (\sigma_1(s) \times \ldots \times \sigma_r(s))^{\frac{1}{r}} = \mathrm{Nm}(s)^{\frac{1}{r}}.$$

The equality holds when $\sigma_1(s) = \cdots = \sigma_r(s)$. In such case $\sigma_1^2(s) + \ldots + \sigma_r^2(s) = r\sigma_1^2(s) = r(\mathrm{Nm}(s))^{\frac{2}{r}}$. As $\mathrm{Nm}(s) \in \mathbb{N}$, the minimum value of $r(\mathrm{Nm}(s))^{\frac{2}{r}} = r$, which happens when $\mathrm{Nm}(s) = 1$ and $\sigma_1^2(s) = 1$ implying $\sigma_1(s) = 1$. Thus $\delta_1 = \sqrt{r}$ and on applying Lemma 10 we have $A_1 = \{\sigma(1)\}$. □

Lemma 12. *Let $\mathfrak{b}_1\mathfrak{N}$ be an ideal in $\mathcal{O}$ such that $\mathfrak{b}_1\mathfrak{N} = \tilde{s}\mathcal{O}$ with $\tilde{s} \in \mathbb{Z}$. Then, $\delta_1 = |\tilde{s}|\sqrt{r}$ and $A_1 = \{\sigma(|\tilde{s}|)\}$.*

Proof. We wish to minimize $\inf\{\|\sigma(s)\| : s \in \mathfrak{b}_1\mathfrak{N}/\pm, s \neq 0\}$. Without loss of generality this is equivalent to minimizing $\|\sigma(s)\|^2 = \sigma_1^2(s) + \ldots + \sigma_r^2(s)$ on the set

$$B = \{s \in \mathfrak{b}_1\mathfrak{N}/\pm : s \neq 0, \sigma_j(s) \geq 0 \text{ for } j = 1, .., r\}.$$

Proceeding similar to the argument in Lemma 11, the minimum possible value of $\|\sigma(s)\|^2$ is $r(\mathrm{Nm}(s))^{\frac{2}{r}}$. For $s \in B$, there exists $s' \in \mathcal{O}$ such that $s = \tilde{s}s'$. Since norm is multiplicative, the minimum value for $r(\mathrm{Nm}(s))^{\frac{2}{r}}$ is attained at $\tilde{s}$. This shows $\|\sigma(s)\|^2$ attains its minimum value $r(\mathrm{Nm}(\tilde{s}))^{\frac{2}{r}}$ at $\sigma(\tilde{s}) = (|\tilde{s}|, \ldots, |\tilde{s}|)$. By Lemma 10, we get $A_1 = \{\sigma(|\tilde{s}|)\}$. □

Remark 5. It must be noted that Lemma 12 is a sufficient condition to have $|A_1| = 1$. Therefore in the hypothesis of Theorem 2, the condition that $\mathfrak{b}_1\mathfrak{N} = \tilde{s}\mathcal{O}$ with $\tilde{s} \in \mathbb{Z}$ can be replaced with the condition that $|\{s \in \mathfrak{b}_1\mathfrak{N}/\pm : s \neq 0, \|\sigma(s)\| = \delta_1\}| = 1$.

We now consider a sufficient condition for the non-vanishing of certain Kloosterman sums that helps us to prove Theorem 3.

Lemma 13. *Let $\mathfrak{N} = \tilde{s}\mathcal{O}$ be unramified with $\tilde{s} \in \mathbb{N}$, $\tilde{s}$ be squarefree. Let $\tilde{m}_1, \tilde{m}_2 \in \mathfrak{d}_+^{-1}$ and $\omega_{\mathfrak{N}}$ be trivial. Then*

$$S_{\omega_{\mathfrak{N}}}(\tilde{m}_1, \tilde{m}_2; 1; \tilde{s}) \neq 0.$$

Proof. We have

$$S_{\omega_{\mathfrak{N}}}(\tilde{m}_1, \tilde{m}_2; 1; \tilde{s}) = \prod_{\nu < \infty} S_{\omega_{\mathfrak{N},\nu}}(\tilde{m}_{1\nu}, \tilde{m}_{2\nu}; 1; \tilde{s}_\nu).$$

Let $\tilde{s}\mathcal{O} = \prod_{l=1}^{t'} \mathfrak{p}_l$ for distinct prime ideals $\mathfrak{p}_l$. Let ν_l be the corresponding valuation for the prime ideal $\mathfrak{p}_l$. Hence

$$\prod_{\nu < \infty} S_{\omega_{\mathfrak{N},\nu}}(\tilde{m}_{1\nu}, \tilde{m}_{2\nu}; 1; \tilde{s}_\nu) = \prod_{l=1}^{t'} S_{\omega_{\mathfrak{N},\nu_l}}(m_{1\nu_l}, m_{2\nu_l}; 1; \varpi_{\nu_l})$$

where ϖ_{ν_l} is a generator of the maximal ideal $(\mathfrak{p}_l)_{\nu_l} = \mathfrak{p}_l\mathcal{O}_{\nu_l}$ and $m_{1\nu_l}, m_{2\nu_l} \in \hat{\mathfrak{d}}_{\nu_l}^{-1}$. For trivial $\omega_{\mathfrak{N}}$, we have

$$\begin{aligned} S_{\omega_{\mathfrak{N},\nu_l}}(m_{1\nu_l}, m_{2\nu_l}; 1; \varpi_{\nu_l}) &= \sum_{\substack{s_1, s_2 \in \mathcal{O}_{\nu_l}/\varpi_{\nu_l}\mathcal{O}_{\nu_l} \\ s_1 s_2 \equiv 1 \pmod{\varpi_{\nu_l}\mathcal{O}_{\nu_l}}}} \theta_{\nu_l}\Big(\frac{m_{1\nu_l}s_1 + m_{2\nu_l}s_2}{\varpi_{\nu_l}}\Big) \\ &= \sum_{\substack{s_1, s_2 \in \mathcal{O}_{\nu_l}/\varpi_{\nu_l}\mathcal{O}_{\nu_l} \\ s_1 s_2 \equiv 1 \pmod{\varpi_{\nu_l}\mathcal{O}_{\nu_l}}}} e\left(\mathrm{Tr}\Big(\frac{m_{1\nu_l}s_1 + m_{2\nu_l}s_2}{\varpi_{\nu_l}}\Big)\right), \end{aligned}$$

where $e(x)=e^{2\pi ix}$. Let $p_l=\mathfrak{p}_l\cap\mathbb{Z}$, and we have

$$\begin{aligned}\left[e\left(\mathrm{Tr}\Big(\frac{m_{1\nu_l}s_1+m_{2\nu_l}s_2}{\varpi_{\nu_l}}\Big)\right)\right]^{p_l}&=e\left(\mathrm{Tr}\Big(p_l\cdot\frac{m_{1\nu_l}s_1+m_{2\nu_l}s_2}{\varpi_{\nu_l}}\Big)\right)\\&=e\left(\mathrm{Tr}\Big(\frac{m_{1\nu_l}p_ls_1}{\varpi_{\nu_l}}\Big)\right)\cdot e\left(\mathrm{Tr}\Big(\frac{m_{2\nu_l}p_ls_2}{\varpi_{\nu_l}}\Big)\right)\\&=1.\end{aligned}$$

The last equality follows from the fact that $\frac{m_{1\nu_l}p_ls_1}{\varpi_{\nu_l}}$ and $\frac{m_{2\nu_l}p_ls_2}{\varpi_{\nu_l}}$ belong to the local inverse different $\hat{\mathfrak{d}}_{\nu_l}^{-1}$.

Suppose $S_{\omega_{\mathfrak{N},\nu_l}}(m_{1\nu_l},m_{2\nu_l};1;\varpi_{\nu_l})=0$. Note that $\mathbb{Z}\Big[e^{\frac{2\pi i}{p_l}}\Big]$ is isomorphic to $\mathbb{Z}[x]/(\Phi_{p_l}(x))$ where $e^{\frac{2\pi i}{p_l}}$ gets mapped to $x+(\Phi_{p_l}(x))$ and $\Phi_{p_l}(x)=1+x+x^2+\cdots+x^{p_l-1}$. We further have $\mathbb{Z}[x]/(\Phi_{p_l}(x),p_l)=\mathbb{Z}[x]/((x-1)^{p_l-1},p_l)$. Consider the ring homomorphism

$$\mathbb{Z}[x]/(\Phi_{p_l}(x))\to\mathbb{Z}[x]/((x-1)^{p_l-1},p_l)\to\mathbb{Z}[x]/((x-1),p_l)\to\mathbb{F}_{p_l},$$

where

$$\begin{aligned}x+(\Phi_{p_l}(x))&\mapsto x+((x-1)^{p_l-1},p_l)\mapsto x+((x-1),p_l)\\&=1+[x-1]+((x-1),p_l)=1+((x-1),p_l)\mapsto 1.\end{aligned}$$

This implies that $e^{\frac{2\pi i}{p_l}}$ gets mapped to 1 via the ring homomorphism. Now $e\left(\mathrm{Tr}\Big(\frac{m_{1\nu_l}s_1+m_{2\nu_l}s_2}{\varpi_{\nu_l}}\Big)\right)$ lies in $\mathbb{Z}\Big[e^{\frac{2\pi i}{p_l}}\Big]$ so that $e\left(\mathrm{Tr}\Big(\frac{m_{1\nu_l}s_1+m_{2\nu_l}s_2}{\varpi_{\nu_l}}\Big)\right)$ gets mapped to 1 via the ring homomorphism. This implies $S_{\omega_{\mathfrak{N},\nu_l}}(m_{1\nu_l},m_{2\nu_l};1;\varpi_{\nu_l})$ will get mapped to $p_l^f-1=-1\in\mathbb{F}_{p_l}$, where $\big|\mathcal{O}_{\nu_l}/\varpi_{\nu_l}\mathcal{O}_{\nu_l}\big|=p_l^f$ for some natural number f. This is a contradiction since image of $S_{\omega_{\mathfrak{N},\nu_l}}(m_{1\nu_l},m_{2\nu_l};1;\varpi_{\nu_l})$ should be 0. □

Corollary 14. *Let F have odd narrow class number and suppose that the assumptions of Theorem 1 hold true. Let $\mathfrak{b}_1\mathfrak{N}=\tilde{s}\mathcal{O}$ with $\tilde{s}\in\mathbb{Z}$. As $k_0\to\infty$, assume that $S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_1\mathfrak{b}_1^{-2};s\mathfrak{b}_1^{-1})\neq 0$ for all m_1 and m_2. Then*

$$\begin{aligned}&\left|\frac{e^{2\pi tr_{\mathbb{Q}}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\left[\prod_{j=1}^r\frac{(k_j-2)!}{(4\pi\sqrt{\sigma_j(m_1m_2)})^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{n}}^{\phi}W_{m_1}^{\phi}(1)\overline{W_{m_2}^{\phi}(1)}}{\|\phi\|^2}-\hat{T}(m_1,m_2,\mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_{\mathfrak{N}}(m_1/s)\omega_{\mathfrak{f}}(s)}\right|\\&\qquad\gg\prod_{j=1}^r(k_j-1)^{-\frac{1}{3}}\end{aligned}$$

Proof. Using Theorem 1, Lemma 9 and Lemma 12, we have

$$\begin{aligned}&\frac{e^{2\pi tr_{\mathbb{Q}}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\left[\prod_{j=1}^r\frac{(k_j-2)!}{(4\pi\sqrt{\sigma_j(m_1m_2)})^{k_j-1}}\right]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{n}}^{\phi}W_{m_1}^{\phi}(1)\overline{W_{m_2}^{\phi}(1)}}{\|\phi\|^2}-\hat{T}(m_1,m_2,\mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_{\mathfrak{N}}(m_1/s)\omega_{\mathfrak{f}}(s)}\\&=\omega_{\mathfrak{f}}(s\mathfrak{b}_1^{-1})S_{\omega_{\mathfrak{N}}}(m_1,m_2;\eta_1\mathfrak{b}_1^{-2};s\mathfrak{b}_1^{-1})\frac{\sqrt{\mathrm{Nm}(\eta_1)}}{\mathrm{Nm}(s)}\times\prod_{j=1}^r\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_1m_1m_2)}}{|\sigma_j(s)|}\Big)\\&\quad+\mathrm{o}\Big(\prod_{j=1}^r\big(k_j-1\big)^{-\frac{1}{3}}\Big).\end{aligned}$$

Note that

$$\frac{4\pi\sqrt{\sigma_j(\eta_1m_1m_2)}}{|\sigma_j(s)|}=\frac{4\pi\sqrt{\sigma_j(\eta_1m_1m_2)}}{(\delta_1/\sqrt{r})}=\frac{2\pi\sqrt{\sigma_j(\eta_1m_1m_2)}}{\delta}.$$

By the given condition

$$\frac{2\pi\gamma_j\sqrt{\sigma_j(m_1m_2)}}{\delta}\in\Big((k_j-1)-(k_j-1)^{\frac{1}{3}},\big(k_j-1\big)\Big),$$

we can write

$$\frac{2\pi\gamma_j\sqrt{\sigma_j(m_1m_2)}}{\delta}=(k_j-1)+d(k_j-1)^{\frac{1}{3}}$$

with $d \in (-1, 0)$. Using Lemma 8, part (iv) we have

$$\prod_{j=1}^{r} J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_1 m_1 m_2)}}{|\sigma_j(s)|}\Big) \gg \prod_{j=1}^{r}(k_j-1)^{-\frac{1}{3}}$$

and this completes the proof of this corollary. □

Proof of Theorem 2. The theorem follows from Lemmas 9, 12 and Corollary 14. □

Remark 6. Note that the assumption $S_{\omega_{\mathfrak{N}}}(m_1, m_2; \eta_1 \mathfrak{b}_1^{-2}; s\mathfrak{b}_1^{-1}) \neq 0$ is essential to have a lower bound like Corollary 14. For instance the main term of Theorem 1.7 of [JS20] is

$$J_{k-1}(4\pi\sqrt{mn})\frac{\mu(N)}{N}\prod_{p|N}(1-p^{-2}).$$

This has a lower bound of $\frac{1}{k^{\frac{1}{3}}}$ for squarefree levels as we have $\mu(N) \neq 0$ if and only if N is squarefree. Hence, the non-vanishing of the Kloosterman sum involved plays an important role.

5. Lower bounds for discrepancy between measures: Proof of Theorem 3

In order to prove Theorem 3, we first obtain a result similar to Corollary 14 under the assumption that

$$\frac{2\pi\gamma_j\sqrt{\sigma_j(m_1m_2)}}{\delta} \in \Big((k_j-1)-(k_j-1)^{\frac{1}{3}}, (k_j-1)\Big),$$

for all j, where $\gamma_j = \frac{\sqrt{\sigma_j(\eta_1)}}{|\sigma_j(d)|}$.

Lemma 15. *Let F have odd narrow class number, $\mathfrak{b}_1\mathfrak{N} = \tilde{s}\mathcal{O}$ with $\tilde{s} \in \mathbb{Z}$ and $\mathfrak{d} = d\mathcal{O}$. As $k_0 \to \infty$, for $k = (k_1, \ldots, k_r)$, suppose $\frac{2\pi\gamma_j\sqrt{\sigma_j(m_1m_2)}}{\delta} \in \Big((k_j-1)-(k_j-1)^{\frac{1}{3}}, (k_j-1)\Big)$ for all j, where $\gamma_j = \frac{\sqrt{\sigma_j(\eta_1)}}{|\sigma_j(d)|}$. Further assume that $S_{\omega_{\mathfrak{N}}}(m_1, m_2; \eta_1\mathfrak{b}_1^{-2}; s\mathfrak{b}_1^{-1}) \neq 0$ for a sequence of m_1 and m_2. Then*

$$\Bigg|\frac{e^{2\pi tr_{\mathbb{Q}}^F(m_1+m_2)}}{\psi(\mathfrak{N})}\Bigg[\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi\sqrt{\sigma_j(m_1m_2)})^{k_j-1}}\Bigg]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{n}}^{\phi}W_{m_1}^{\phi}(1)\overline{W_{m_2}^{\phi}(1)}}{\|\phi\|^2}$$

$$-\hat{T}(m_1, m_2, \mathfrak{n})\frac{\sqrt{d_F\mathrm{Nm}(\mathfrak{n})}}{\omega_{\mathfrak{N}}(m_1/s)\omega_{\mathrm{f}}(s)}\Bigg| \gg \prod_{j=1}^{r}(k_j-1)^{-\frac{1}{3}}$$

Proof. To see this, change $\gamma_j = \frac{\sqrt{\sigma_j(\eta_1)}}{|\sigma_j(d)|}$ in the proofs of Theorem 1 and Theorem 2 in place of $\gamma_j = \sqrt{\sigma_j(\eta_1)}$. The proof follows similarly. □

Corollary 16. *Let F has narrow class number equal to 1, $\mathfrak{p} = p\mathcal{O}$ and $\mathfrak{d} = d\mathcal{O}$. Let $\mathfrak{b}_1\mathfrak{N} = \tilde{s}\mathcal{O}$ with $\tilde{s} \in \mathbb{Z}$ and $|\tilde{s}|$ being squarefree. Further, let $\omega_{\mathfrak{N}}$ be trivial and ℓ be odd such that*

$$\frac{2\pi\gamma_j\sqrt{\sigma_j(p^\ell)}}{\delta|\sigma_j(d)|} \in \Big((k_j-1)-(k_j-1)^{\frac{1}{3}}, (k_j-1)\Big)$$

for all j, where $\gamma_j = \frac{\sqrt{\sigma_j(\eta_1)}}{|\sigma_j(d)|}$. Then

$$\Bigg|\frac{1}{\sqrt{\mathrm{Nm}(p^\ell)}}\Bigg[\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi)^{k_j-1}}\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{p}^\ell}^{\phi}}{\|\phi\|^2}\Bigg]\Bigg| \gg \prod_{j=1}^{r}(k_j-1)^{-\frac{1}{3}}.$$

Proof. Let us take $m_1 = \frac{p^\ell}{d}$ with ℓ odd and $m_2 = \frac{1}{d}$ in Corollary 6 to get

$$\frac{e^{4\pi r}\mathrm{Nm}(d)}{\psi(\mathfrak{N})d_F^2\sqrt{\mathrm{Nm}(p^\ell)}}\Bigg[\prod_{j=1}^{r}\frac{(k_j-2)!}{(4\pi)^{k_j-1}}\Bigg]\sum_{\phi\in\mathcal{F}}\frac{\lambda_{\mathfrak{p}^l}^{\phi}}{\|\phi\|^2} = \hat{T}\Big(\frac{p^\ell}{d}, \frac{1}{d}, \mathcal{O}\Big)\frac{\sqrt{d_F}}{\omega_{\mathfrak{N}}\Big(\frac{p^\ell}{sd}\Big)\omega_{\mathrm{f}}(s)}$$

$$+\sum_{s\in\mathfrak{b}_1\mathfrak{N}/\pm, s\neq 0}\Bigg\{\omega_{\mathrm{f}}(s\mathfrak{b}_1^{-1})S_{\omega_{\mathfrak{N}}}\Big(\frac{p^\ell}{d}, \frac{1}{d}; 1; \tilde{s}\Big)\times\frac{\sqrt{\mathrm{Nm}(\eta_1)}}{\mathrm{Nm}(s)}\times\prod_{j=1}^{r}\frac{2\pi}{(\sqrt{-1})^{k_j}}J_{k_j-1}\Big(\frac{4\pi\sqrt{\sigma_j(\eta_1 p^\ell)}}{|\sigma_j(sd)|}\Big)\Bigg\}.$$

Since ℓ is odd, $\hat{T}\left(\frac{p^\ell}{d}, \frac{1}{d}, \mathcal{O}\right) = 0$ and Lemma 13 implies that $S_{\omega_{\mathfrak{N}}}\left(\frac{p^\ell}{d}, \frac{1}{d}; 1; \tilde{s}\right) \neq 0$. Applying Lemma 15, we get

$$\left| \frac{e^{4\pi r}\mathrm{Nm}(d)}{\psi(\mathfrak{N})d_F^2\sqrt{\mathrm{Nm}(p^\ell)}} \left[\prod_{j=1}^{r} \frac{(k_j-2)!}{(4\pi)^{k_j-1}}\right] \sum_{\phi\in\mathcal{F}} \frac{\lambda_{\mathfrak{p}^\ell}^{\phi}}{\|\phi\|^2} \right| \gg \prod_{j=1}^{r}(k_j-1)^{-\frac{1}{3}}.$$

Thus

$$\left| \frac{1}{\sqrt{\mathrm{Nm}(p^\ell)}} \left[\prod_{j=1}^{r} \frac{(k_j-2)!}{(4\pi)^{k_j-1}} \sum_{\phi\in\mathcal{F}} \frac{\lambda_{\mathfrak{p}^\ell}^{\phi}}{\|\phi\|^2}\right] \right| \gg \prod_{j=1}^{r}(k_j-1)^{-\frac{1}{3}}.$$

□

Consider $\kappa_{\mathfrak{p}^\ell}^{\phi} = \frac{\lambda_{\mathfrak{p}^\ell}^{\phi}}{\sqrt{\mathrm{Nm}(\mathfrak{p}^\ell)}}$ and the discrete measure

$$\tilde{\nu}_{k,\mathfrak{N}} := \prod_{j=1}^{r} \frac{(4\pi)^{k_j-1}}{(k_j-2)!} \sum_{\phi\in\mathcal{F}} \frac{\delta_{\kappa_{\mathfrak{p}}^{\phi}}}{\|\phi\|^2}.$$

We now prove Theorem 3.

Proof of Theorem 3. Let $k_{lj} = \left[\frac{2\pi\gamma_j\sqrt{\sigma_j(p^l)}}{\delta|\sigma_j(d)|}\right] - 1$, where $[x]$ denotes the greatest integer part of x. It is easy to check that this k_{lj} satisfies $\frac{2\pi\gamma_j\sqrt{\sigma_j(p^l)}}{\delta|\sigma_j(d)|} \in \left((k_{lj}-1)-(k_{lj}-1)^{\frac{1}{3}}, (k_{lj}-1)\right)$. This now satisfies the hypothesis in Corollary 16 and we get

$$\left| \frac{1}{\sqrt{\mathrm{Nm}(p^l)}} \prod_{j=1}^{r} \frac{(k_{lj}-2)!}{(4\pi)^{k_{lj}-1}} \sum_{\phi\in\mathcal{F}} \frac{\lambda_{\mathfrak{p}^l}^{\phi}}{\|\phi\|^2} \right| \gg \prod_{j=1}^{r}(k_{lj}-1)^{-\frac{1}{3}}. \tag{45}$$

We have (see, for example, Proposition 4.5 of [KL08])

$$\kappa_{\mathfrak{p}^l}^{\phi} = X_l(\kappa_{\mathfrak{p}}^{\phi})$$

where $X_l(2\cos\theta) = \frac{\sin(l+1)\theta}{\sin\theta}$ is the degree l Chebyshev polynomial of second kind. Hence

$$\left| \prod_{j=1}^{r} \frac{(k_{l_j}-2)!}{(4\pi)^{k_{l_j}-1}} \sum_{\phi\in\mathcal{F}} \frac{\kappa_{\mathfrak{p}^l}^{\phi}}{\|\phi\|^2} \right| \gg \prod_{j=1}^{r}(k_{l_j}-1)^{-\frac{1}{3}},$$

equivalently,

$$\left| \prod_{j=1}^{r} \frac{(k_{l_j}-2)!}{(4\pi)^{k_{l_j}-1}} \sum_{\phi\in\mathcal{F}} \frac{X_l(\kappa_{\mathfrak{p}}^{\phi})}{\|\phi\|^2} \right| \gg \prod_{j=1}^{r}(k_{l_j}-1)^{-\frac{1}{3}}.$$

Therefore

$$\left| \int_{-2}^{2} X_l(x)\, d\tilde{\nu}_{k_l,\mathfrak{N}}(x) \right| \gg \prod_{j=1}^{r}(k_{l_j}-1)^{-\frac{1}{3}}.$$

The orthogonality of the polynomials $X_l(x)$ with respect to the Sato-Tate measure implies

$$\begin{aligned} \left| \int_{-2}^{2} X_l(x)\, d(\tilde{\nu}_{k_l,\mathfrak{N}} - \mu_\infty)(x) \right| &= \left| \int_{-2}^{2} X_l(x)\, d\tilde{\nu}_{k_l,\mathfrak{N}}(x) - \int_{-2}^{2} X_l(x)\, d\mu_\infty(x) \right| \\ &= \left| \int_{-2}^{2} X_l(x)\, d\tilde{\nu}_{k_l,\mathfrak{N}}(x) \right| \gg \prod_{j=1}^{r}(k_{l_j}-1)^{-\frac{1}{3}}. \end{aligned} \tag{46}$$

Integration by parts and $\left|X_l'(x)\right| \ll l^2$ gives us

$$\left| \int_{-2}^{2} X_l(x)\, d(\tilde{\nu}_{k_l,\mathfrak{N}} - \mu_\infty)(x) \right| \ll l^2 \left| \int_{-2}^{2} d(\tilde{\nu}_{k_l,\mathfrak{N}} - \mu_\infty)(x) \right|. \tag{47}$$

Now

$$\begin{aligned} D(\tilde{\nu}_{k_l,\mathfrak{N}}, \mu_\infty) &\geq \left| \tilde{\nu}_{k_l,\mathfrak{N}}([-2,2]) - \mu_\infty([-2,2]) \right| \\ &= \left| \int_{-2}^{2} d(\tilde{\nu}_{k_l,\mathfrak{N}} - \mu_\infty)(x) \right| \gg \frac{1}{l^2} \left| \int_{-2}^{2} X_l(x)\, d(\tilde{\nu}_{k_l,\mathfrak{N}} - \mu_\infty)(x) \right|, \end{aligned}$$

using equation (47). Equation (46) implies

$$D(\tilde{\nu}_{k_l,\mathfrak{N}},\mu_\infty) \gg \frac{1}{l^2 \times \prod_{j=1}^{r}(k_{l_j}-1)^{\frac{1}{3}}}.$$

However, $k_{l_j} = \left[\frac{2\pi\gamma_j\sqrt{\sigma_j(p^l)}}{|\sigma_j(d)|\delta}\right] - 1$, so $\frac{2\pi\gamma_j\sqrt{\sigma_j(p^l)}}{|\sigma_j(d)|\delta} < 2k_{l_j}$. This implies $\frac{k_{l_j}\delta|\sigma_j(d)|}{\pi\gamma_j} > \left(\sqrt{\sigma_j(p)}\right)^l$ and hence $\log(k_l)_0 \gg l$. Therefore

$$D(\tilde{\nu}_{k_l,\mathfrak{N}},\mu_\infty) \gg \frac{1}{\left(\log(k_l)_0\right)^2 \times \prod_{j=1}^{r}(k_{l_j}-1)^{\frac{1}{3}}},$$

which completes the proof. □

Remark 7. Theorem 3 generalizes Theorem 1.6 of [JS20] to ideals of rings of integers $\mathcal{O}$ with narrow class number 1. In particular, Theorem 3 holds for the space $A_k(\mathcal{O},\omega)$ as $(k_l)_0 \to \infty$ with trivial $\omega_{\mathfrak{N}}$.

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References

[BL84] J.-L. Brylinski and J.-P. Labesse, *Cohomologie d'intersection et fonctions L de certaines variétés de Shimura*, Ann. Sci. École Norm. Sup. (4) **17** (1984), no. 3, 361–412. MR 777375

[Bla06] Don Blasius, *Hilbert modular forms and the Ramanujan conjecture*, Noncommutative geometry and number theory, Aspects Math., E37, Friedr. Vieweg, Wiesbaden, 2006, pp. 35–56. MR 2327298

[BLGG11] T. Barnet-Lamb, T. Gee, and D. Geraghty, *The Sato-Tate conjecture for Hilbert modular forms*, J. Amer. Math. Soc. **24** (2011), no. 2, 411–469.

[BLGHT11] T. Barnet-Lamb, D. Geraghty, M. Harris, and R. Taylor, *A family of Calabi-Yau varieties and potential automorphy II*, Publ. Res. Inst. Math. Sci. **47** (2011), no. 1, 29–98.

[CDF97] B. Conrey, W. Duke, and D. Farmer, *The distribution of the eigenvalues of Hecke operators*, Acta Arith. **78** (1997), no. 4, 405–409.

[Das24] Jishu Das, *A lower bound for the discrepancy in a Sato–Tate type measure*, Ramanujan J. **65** (2024), no. 2, 637 – 658.

[Das26] ———, *Shortest nonzero lattice points in totally real multi-quadratic number fields and applications*, Int. J. Number Theory **22** (2026), no. 6, 1251–1268. MR 5063650

[Del74] Pierre Deligne, *La conjecture de Weil. I*, Inst. Hautes Études Sci. Publ. Math. (1974), no. 43, 273–307. MR 340258

[DS74] Pierre Deligne and Jean-Pierre Serre, *Formes modulaires de poids* 1, Ann. Sci. École Norm. Sup. (4) **7** (1974), 507–530 (1975). MR 379379

[EMP86] H. M. Edgar, R. A. Mollin, and B. L. Peterson, *Class groups, totally positive units, and squares*, Proc. Amer. Math. Soc. **98** (1986), no. 1, 33–37.

[GJS99] Alex Gamburd, Dmitry Jakobson, and Peter Sarnak, *Spectra of elements in the group ring of* SU(2), J. Eur. Math. Soc. (JEMS) **1** (1999), no. 1, 51–85. MR 1677685

[Gol04] E. P. Golubeva, *Distribution of the eigenvalues of Hecke operators*, Zap. Nauchn. Sem. S.-Peterburg. Otdel. Mat. Inst. Steklov. (POMI) **314** (2004), no. Anal. Teor. Chisel i Teor. Funkts. 20, 33–40, 286.

[JS20] J. Jung and N. T. Sardari, *Asymptotic trace formula for the Hecke operators*, Math. Ann. **378** (2020), 513–557.

[KL06] Andrew Knightly and Charles Li, *Traces of Hecke operators*, Mathematical Surveys and Monographs, vol. 133, American Mathematical Society, Providence, RI, 2006. MR 2273356

[KL08] A. Knightly and C. Li, *Petersson's trace formula and the Hecke eigenvalues of Hilbert modular forms*, Modular forms on Schiermonnikoog, Cambridge Univ. Press, Cambridge, 2008, pp. 145–187.

[KL13] ———, *Kuznetsov's trace formula and the Hecke eigenvalues of Maass forms*, Mem. Amer. Math. Soc. **224** (2013), no. 1055, vi+132. MR 3099744

[Lan00] L. J. Landau, *Bessel functions: monotonicity and bounds*, J. London Math. Soc. (2) **61** (2000), no. 1, 197–215.

[Li04] Charles C. C. Li, *Kuznietsov trace formula and weighted distribution of Hecke eigenvalues*, J. Number Theory **104** (2004), no. 1, 177–192. MR 2021634

[Li09] Charles Li, *On the distribution of Satake parameters of* GL_2 *holomorphic cuspidal representations*, Israel J. Math. **169** (2009), 341–373. MR 2460909

[LLW14] Y-K Lau, C. Li, and Y. Wang, *Quantitative analysis of the Satake parameters of* GL_2 *representations with prescribed local representations*, Acta Arith. **164** (2014), no. 4, 355–380.

[LW11] Y-K Lau and Y. Wang, *Quantitative version of the joint distribution of eigenvalues of the Hecke operators*, J. Number Theory **131** (2011), 2262–2281.

[MS09] M. R. Murty and K. Sinha, *Effective equidistribution of eigenvalues of Hecke operators*, J. Number Theory **129** (2009), no. 3, 681–714. MR 2488597

[MS10] ———, *Factoring newparts of Jacobians of certain modular curves*, Proc. Amer. Math. Soc. **138** (2010), no. 10, 3481–3494.

[Ngu05] Lan The Nguyen, *The Ramanujan conjecture for Hilbert modular forms*, ProQuest LLC, Ann Arbor, MI, 2005, Thesis (Ph.D.)–University of California, Los Angeles. MR 2708305

[OM09] Sami Omar and Kamel Mazhouda, *Equirépartition des coefficients de Fourier des fonctions L de carrés et de cubes symétriques*, Ramanujan J. **20** (2009), no. 1, 81–89. MR 2546185

[Roy00] Emmanuel Royer, *Facteurs* $\mathbf{Q}$*-simples de* $J_0(N)$ *de grande dimension et de grand rang*, Bull. Soc. Math. France **128** (2000), no. 2, 219–248. MR 1772442

[Sar87] P. Sarnak, *Statistical properties of eigenvalues of the Hecke operators*, Analytic number theory and Diophantine problems (Stillwater, OK, 1984), Progr. Math., vol. 70, Birkhäuser Boston, Boston, MA, 1987, pp. 321–331.

[Ser97] J-P. Serre, *Répartition asymptotique des valeurs propres de l'opérateur de Hecke* T_p, J. Amer. Math. Soc. **10** (1997), no. 1, 75–102. MR 1396897

[SZ24] Peter Sarnak and Nina Zubrilina, *Convergence to the Plancherel measure of Hecke eigenvalues*, Acta Arith. **214** (2024), 191–213. MR 4772283

[TW16] Hengcai Tang and Yingnan Wang, *Quantitative versions of the joint distributions of Hecke eigenvalues*, J. Number Theory **169** (2016), 295–314. MR 3531241

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